



NTNU – Trondheim
Norwegian University of
Science and Technology

TET4185: Power Markets, Resources and Environment

Environmental Policy

Hossein Farahmand
Professor, IEL, NTNU
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Lecture plan

Learning objectives:

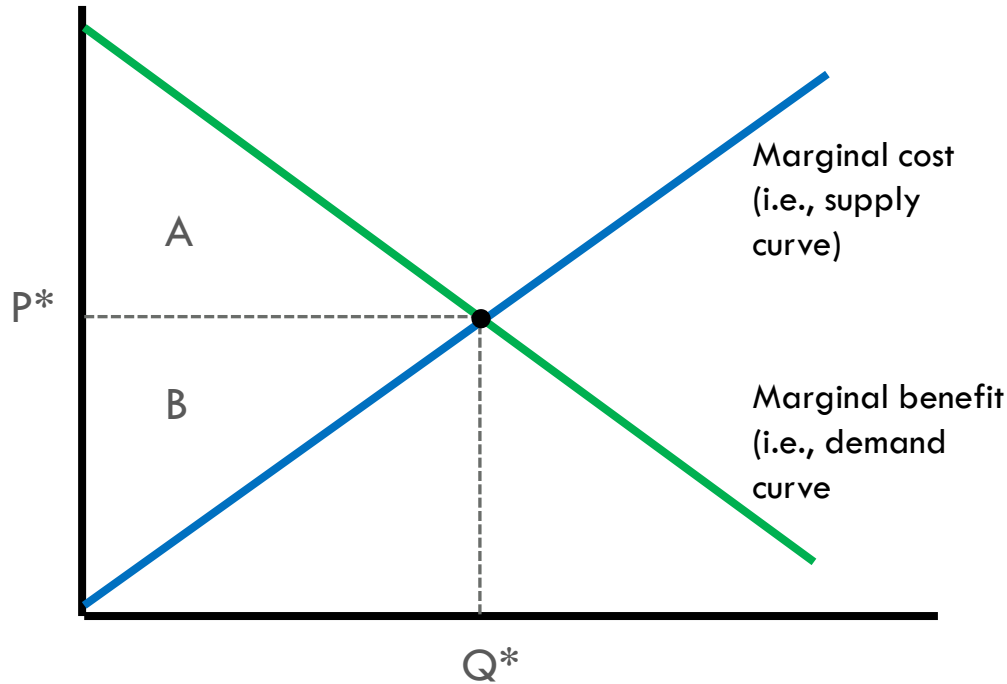
- Understanding fundamental concepts and frameworks in climate policy
- Familiarity with tools for modeling climate policy

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1. Theoretical foundations of environmental policy
2. Climate policy and CO₂ pricing
 - Exercise
3. Cap-and-trade, a.k.a. Emissions Trading Systems
4. The EU Emissions Trading System
5. Technology policy (renewable energy policy)

What is the purpose of environmental policy?

Review: on the case for free markets



A = Consumer Surplus; B = Producer Surplus

First fundamental theorem of welfare economics: The market equilibrium is* **Pareto efficient** (a.k.a. Pareto optimal)

Pareto efficiency: An allocation of resources such that it is impossible to make one person better off without making another worse off (i.e., where it is impossible to make everybody better off)

Important assumptions behind the First Theorem of welfare economics

- There is perfect competition (no market power)
- There is perfect information (no asymmetries between buyers and sellers)
- Agents are rational
- There are no **externalities**

Externalities are defined as:

*“An externality results when the actions of one individual (or firm) have a **direct**, unintentional, and **uncompensated** effect on the well-being of other individuals or the profits of other firms” [1]*

(‘external’ because effects are ‘outside’ of the pricing system)

[1] Keohane and Olmstead. 2007 “Markets and the Environment.” Island Press.

Examples of externalities

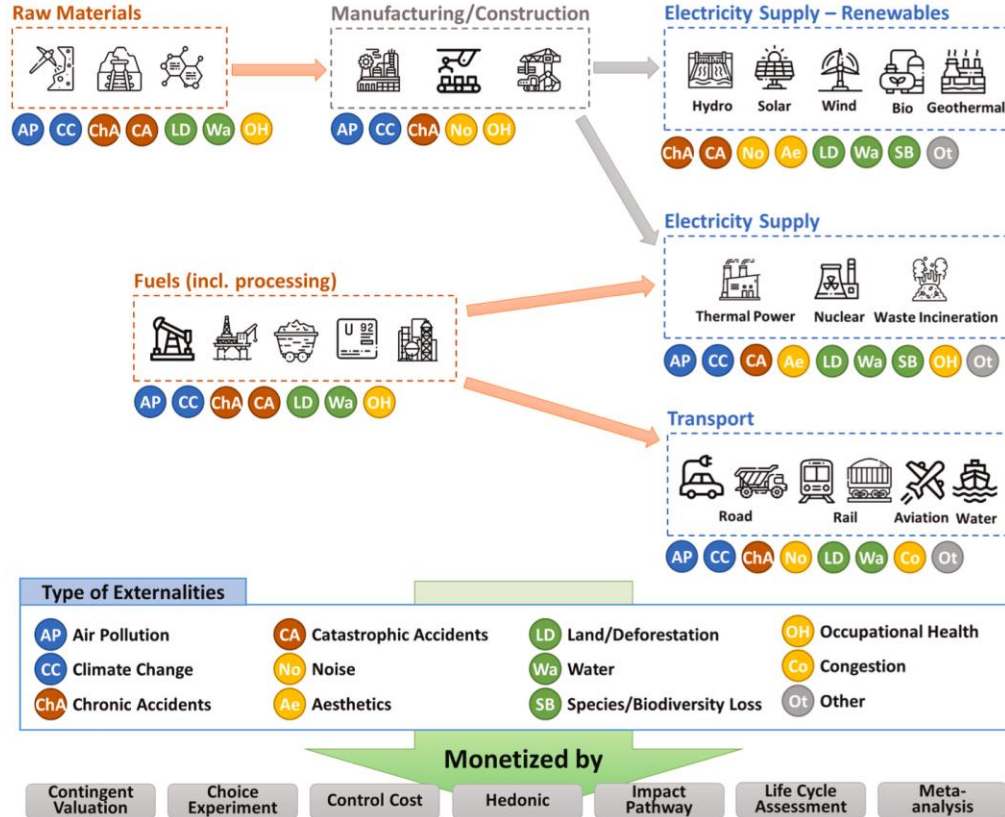
Negative externalities

- **Pollution**
- **Aesthetics:** If my house is an eyesore, it affects my neighbours (Transmission lines, Wind turbines, ...)
- **Use of common resources** (e.g., “tragedy of the commons”)
- **Climate change** (“the mother of all externalities”¹)

Positive externalities

- **Knowledge spill-overs:** knowledge acquired by a firm may end up being used by others, MATPOWER (<https://matpower.org/>)
- **Network externalities:** The more electric vehicles there are, the higher the value of owning an electric vehicle (consider charging infrastructure)

Externalities associated with electricity and transport



How climate change damages are quantified

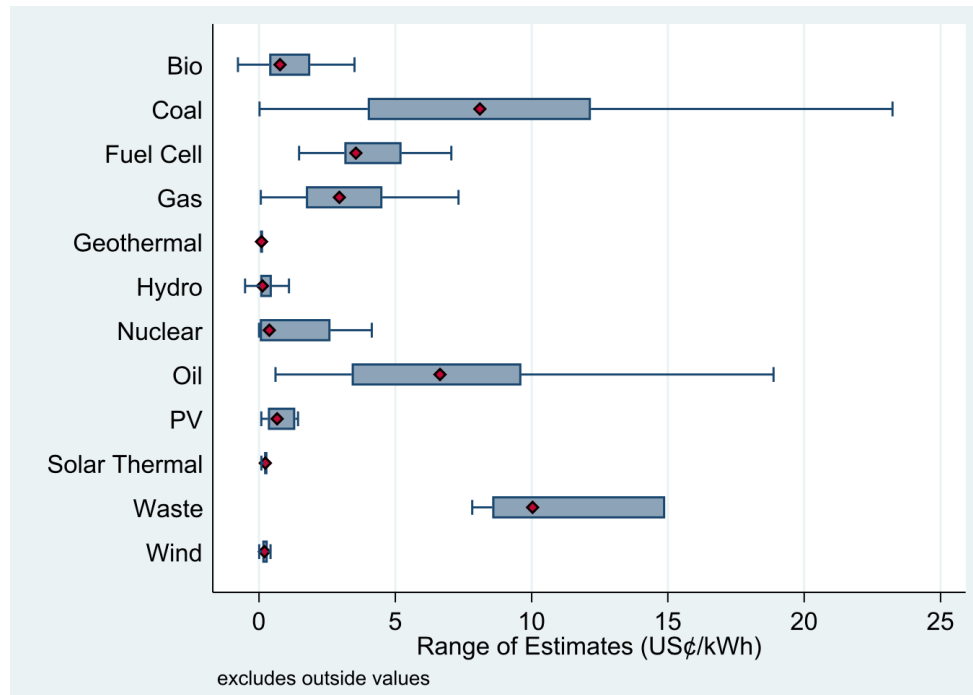
The Social Cost of Carbon (SCC): the economic damage that would result from emitting 1 additional ton of CO₂

The value of the SCC is very difficult to estimate and is subject to a lot of debate.

A recent estimate: \$185/tCO₂ in 2020 (within a range of \$44-413/tCO₂)¹

How significant are electricity-related externalities?

These numbers are only illustrative



The magnitude of negative externalities is in some cases similar to the private cost of production (e.g., for coal)

A big driver is premature mortality:
Deaths related to coal: 57/TWh
Deaths related to nuclear: 0.03/TWh

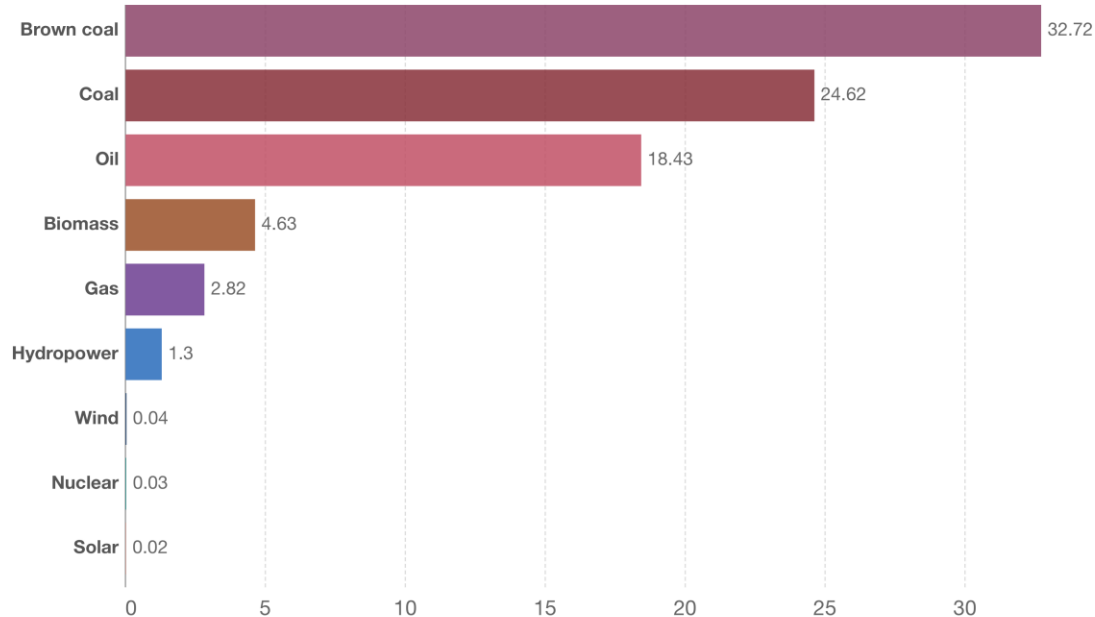
¹ Source: [OWiD](#)

Premature mortalities are a large component of externality estimates

Death rates per unit of electricity production

Death rates are measured based on deaths from accidents and air pollution per terawatt-hour (TWh) of electricity.

Our World
in Data



Source: Markandya & Wilkinson (2007); Sovacool et al. (2016); UNSCEAR (2008; & 2018)

OurWorldInData.org/energy • CC BY

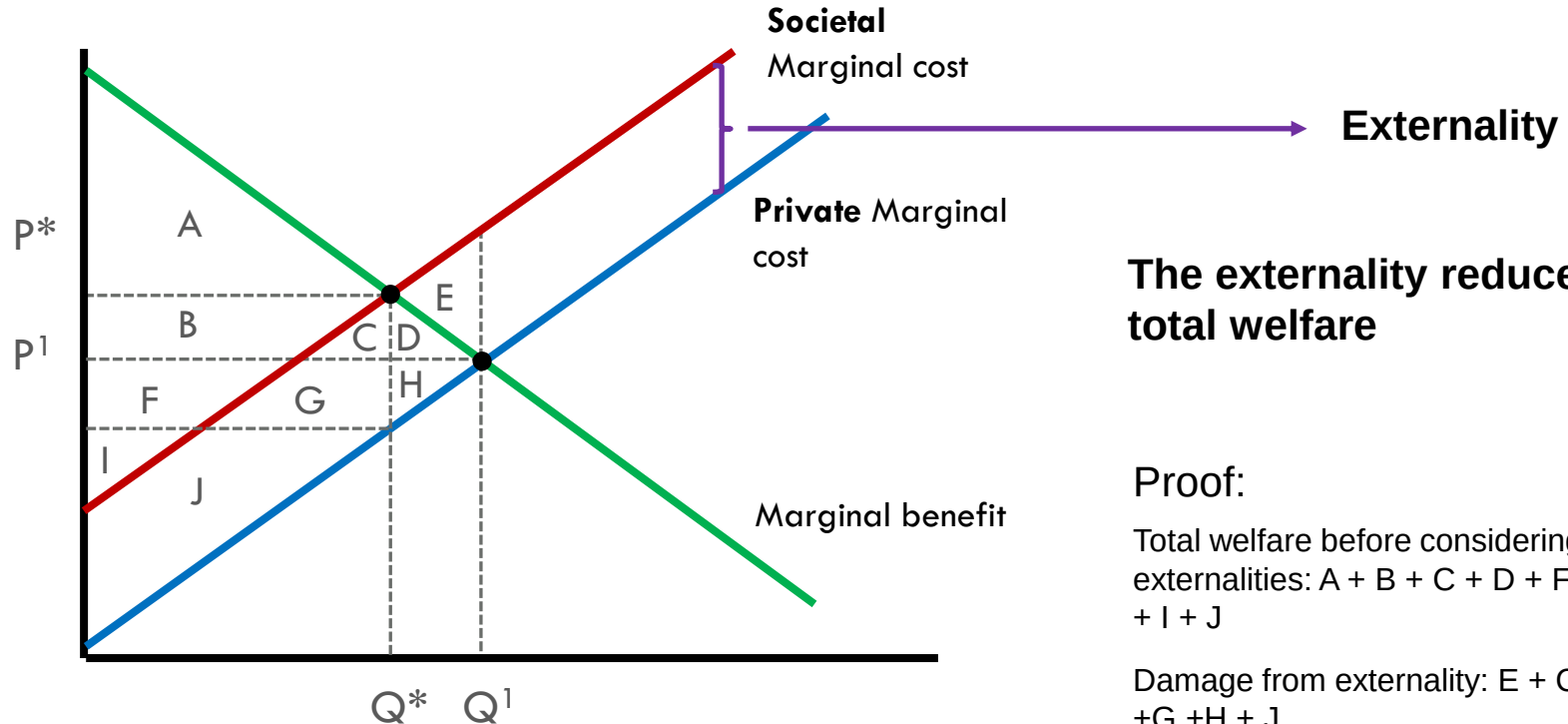


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Source: [OWiD](https://ourworldindata.org/energy)

Externalities in the supply and demand model



The externality reduces total welfare

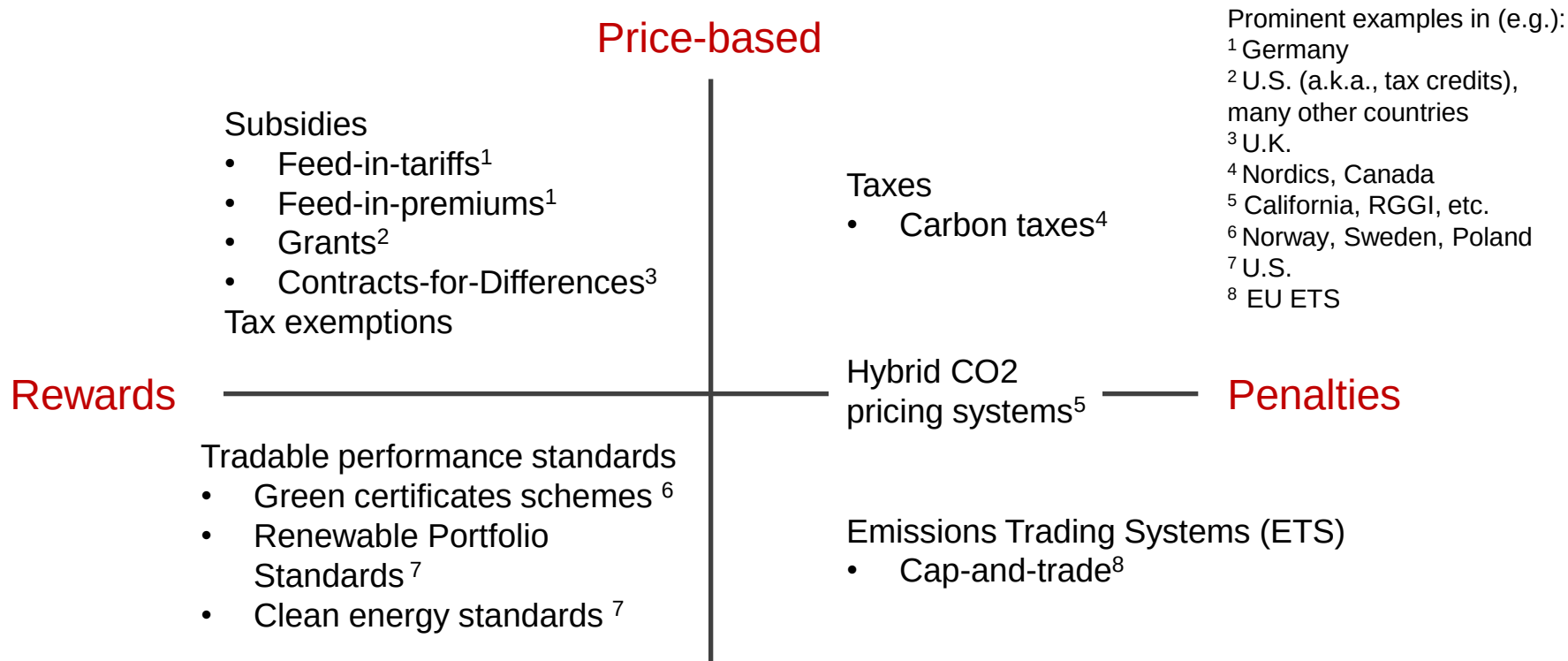
Proof:

Total welfare before considering externalities: $A + B + C + D + F + G + H + I + J$

Damage from externality: $E + C + D + G + H + J$

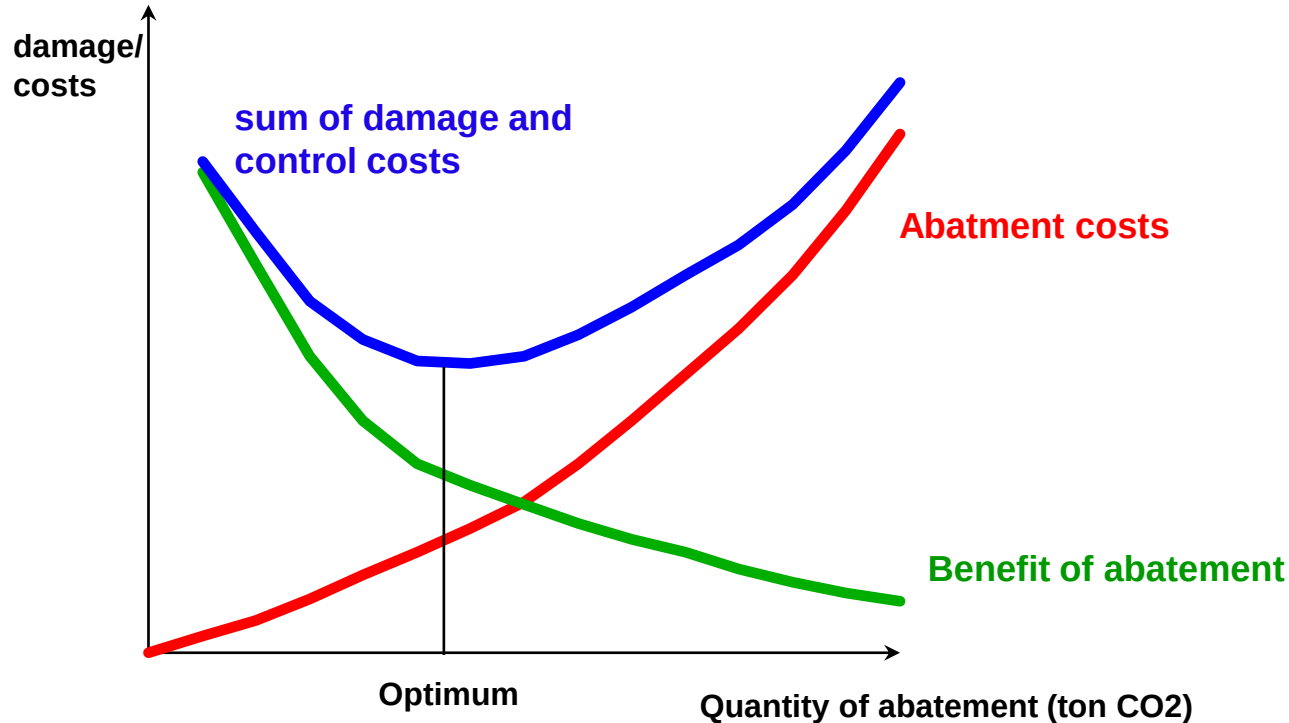
Total welfare with externality: $A + B + F + I - E$

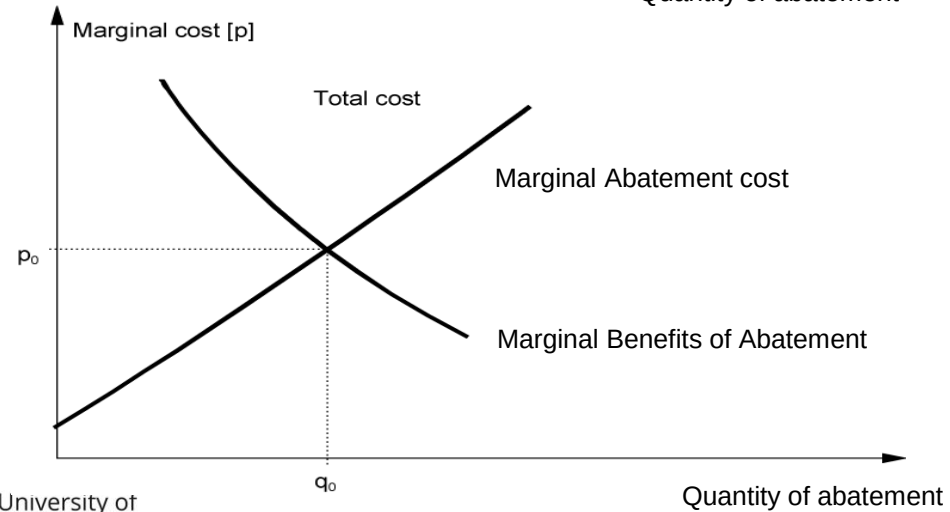
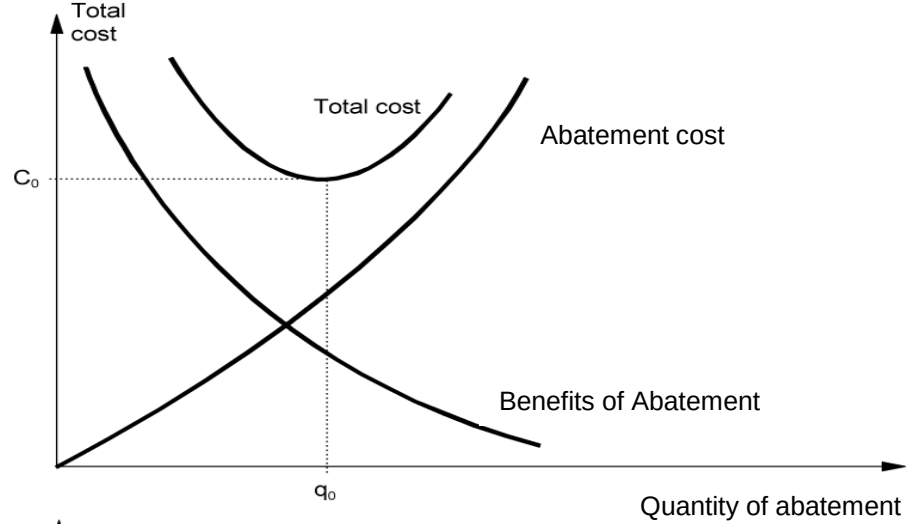
Types of market-based environmental policy instruments implemented around the world



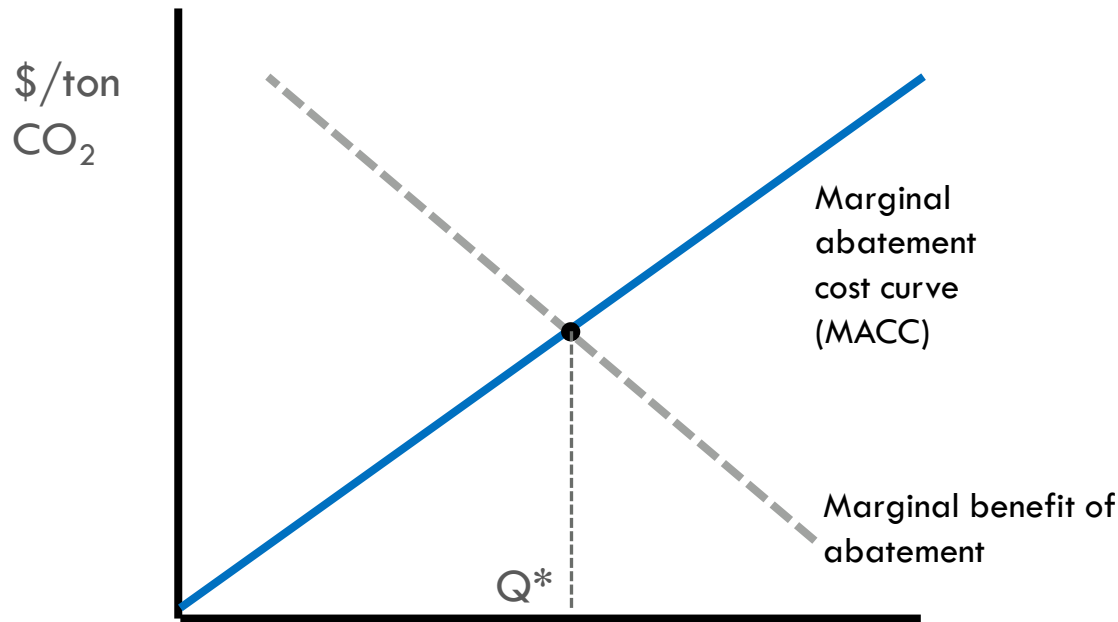
Carbon pricing

Optimal Pollution Control Strategy





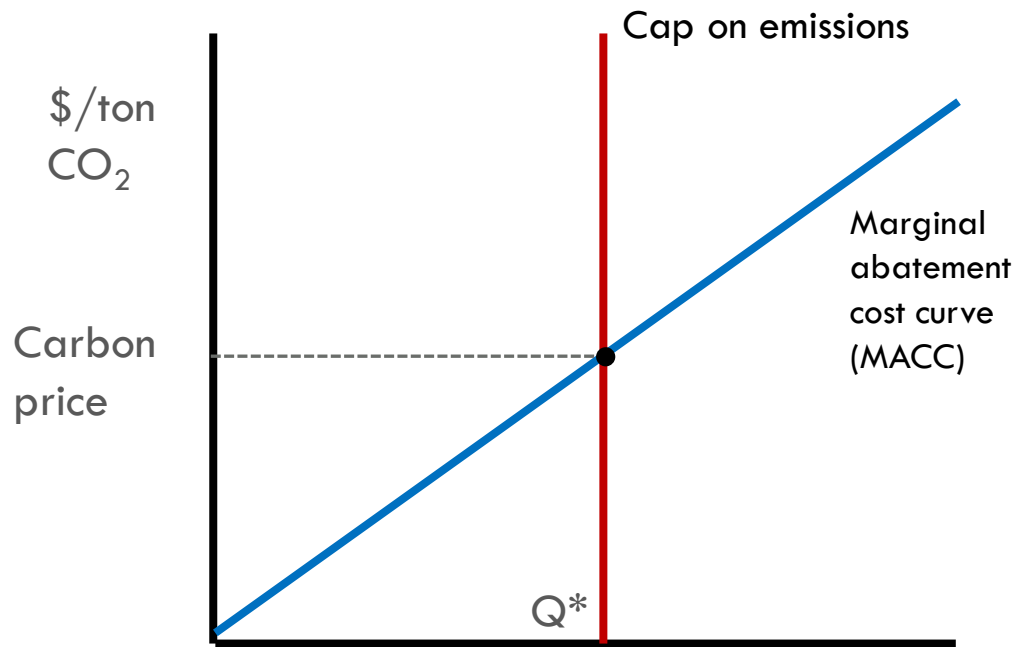
A commonly used tool for understanding climate policy and carbon pricing is the **Marginal Abatement Cost Curve (MACC)**



The **MACC** shows the cost of reducing one additional ton of CO_2

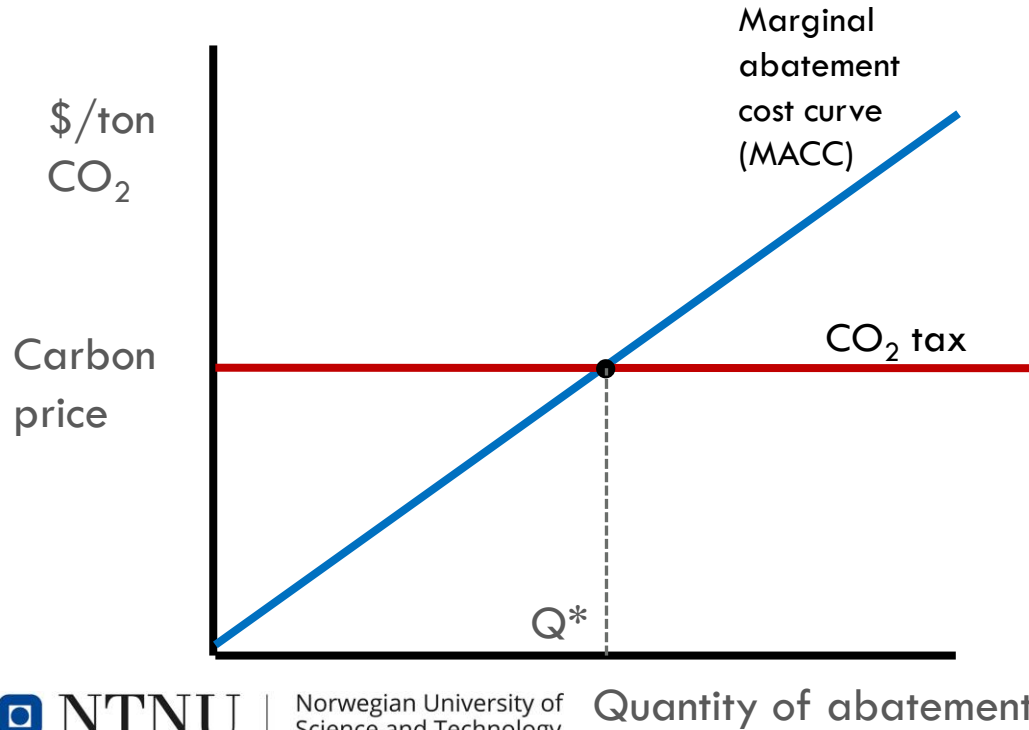
Society derives marginal benefits from abatement, but there is no demand curve without policy

Climate policy can be understood as filling in for the missing demand for abatement



Cap-and-trade (emissions trading systems) impose a cap on emissions

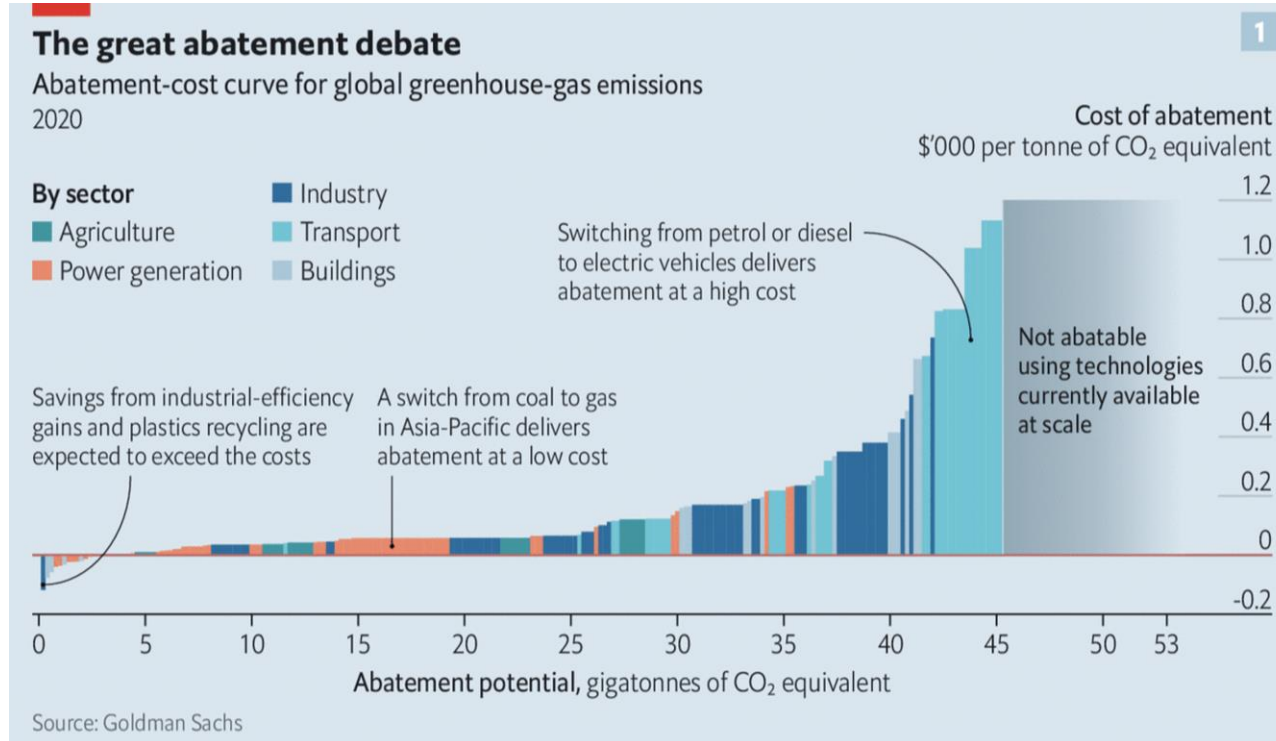
A carbon tax approach



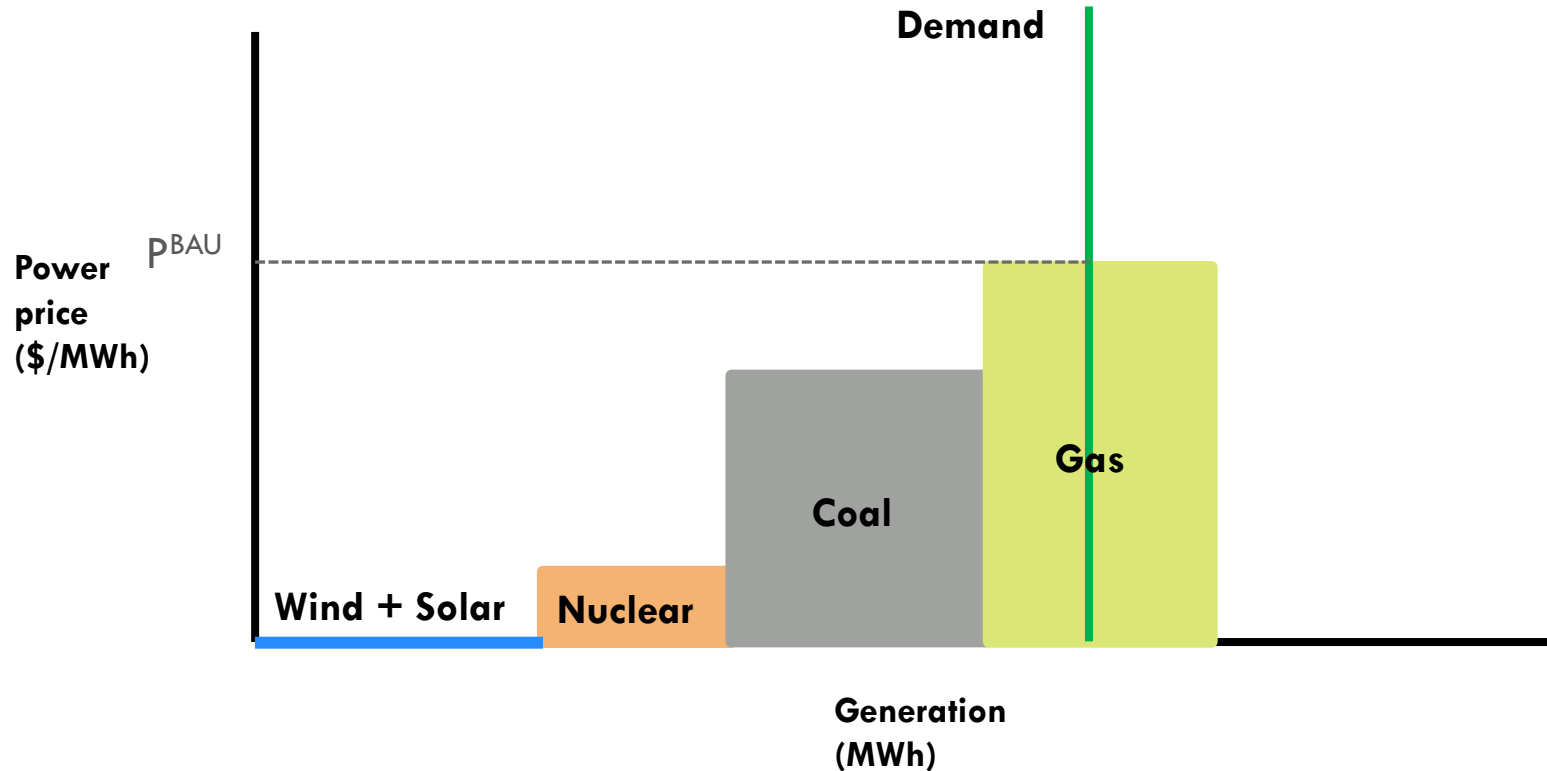
Alternatively, the same objective (i.e., abatement quantity) can (in theory¹) be achieved by an equivalent CO₂ tax

¹ Assuming perfect foresight. Otherwise, see: [Weitzman 1974](#)

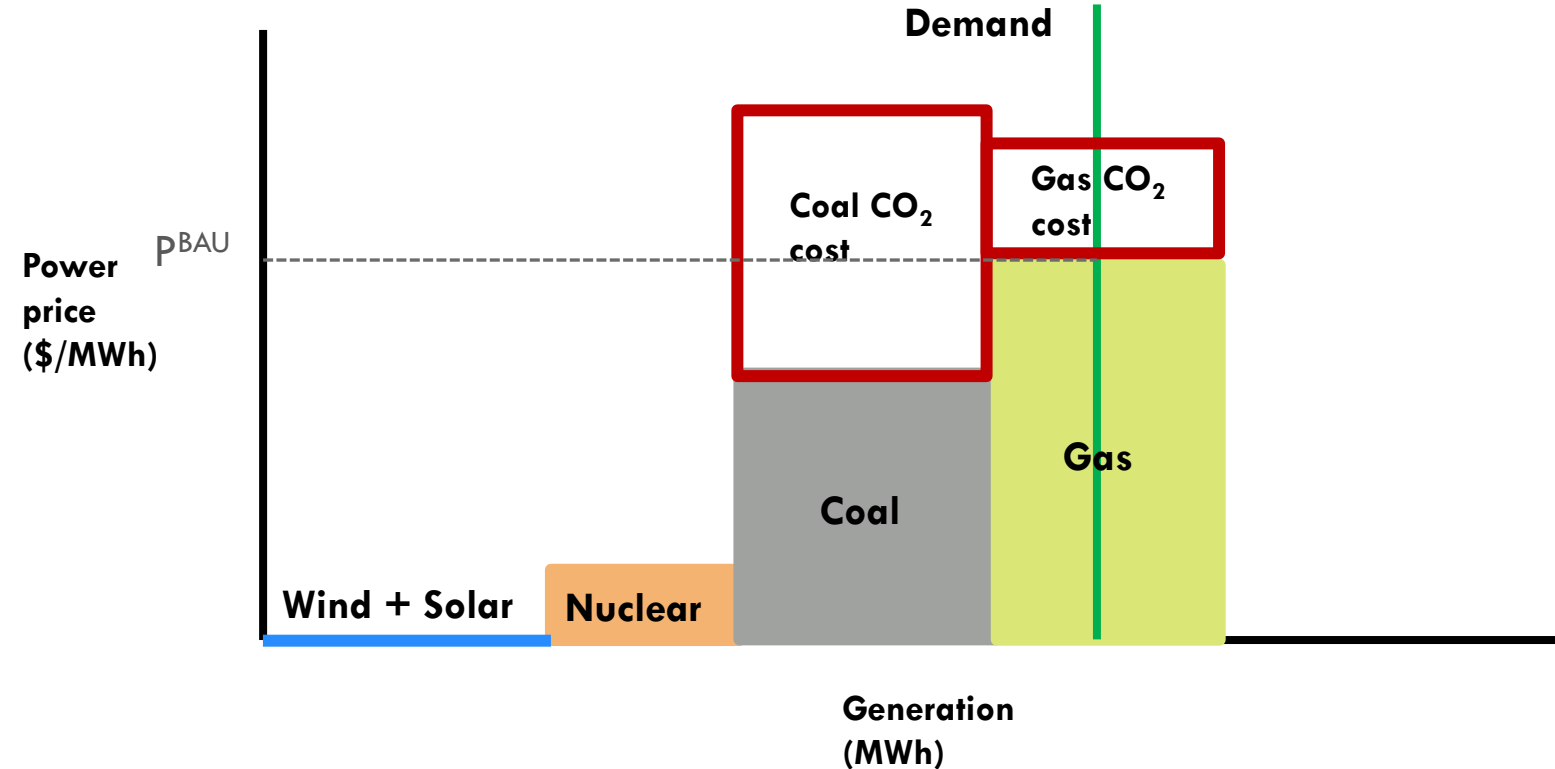
One attempt to quantify Marginal Abatement Costs (values should be treated as illustrative)



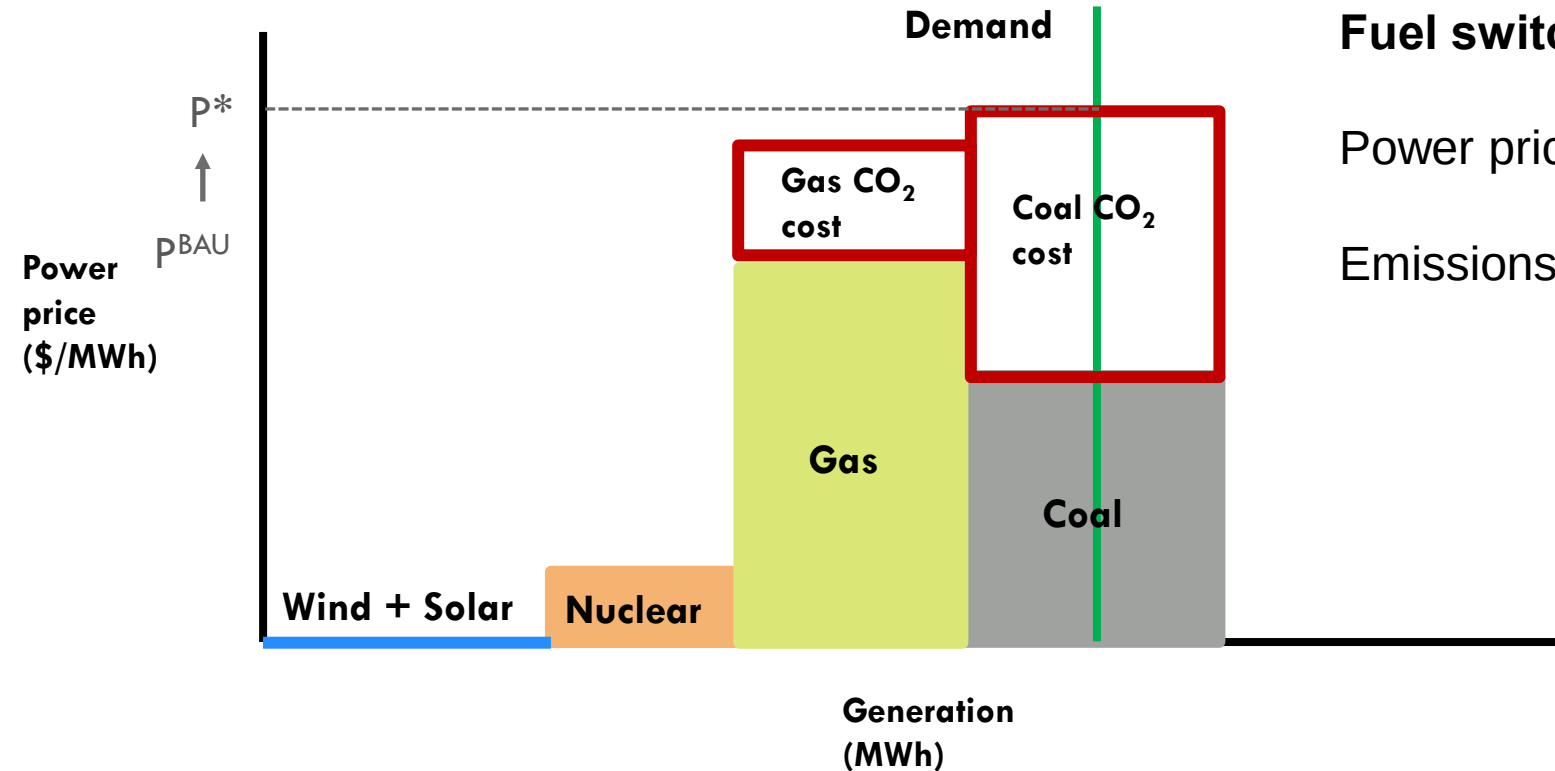
How CO₂ pricing impacts power market clearing



Consider CO₂ costs (intermediate step – the market is not in equilibrium)



The new market equilibrium:



Fuel switching: ...

Power price goes up

Emissions go down

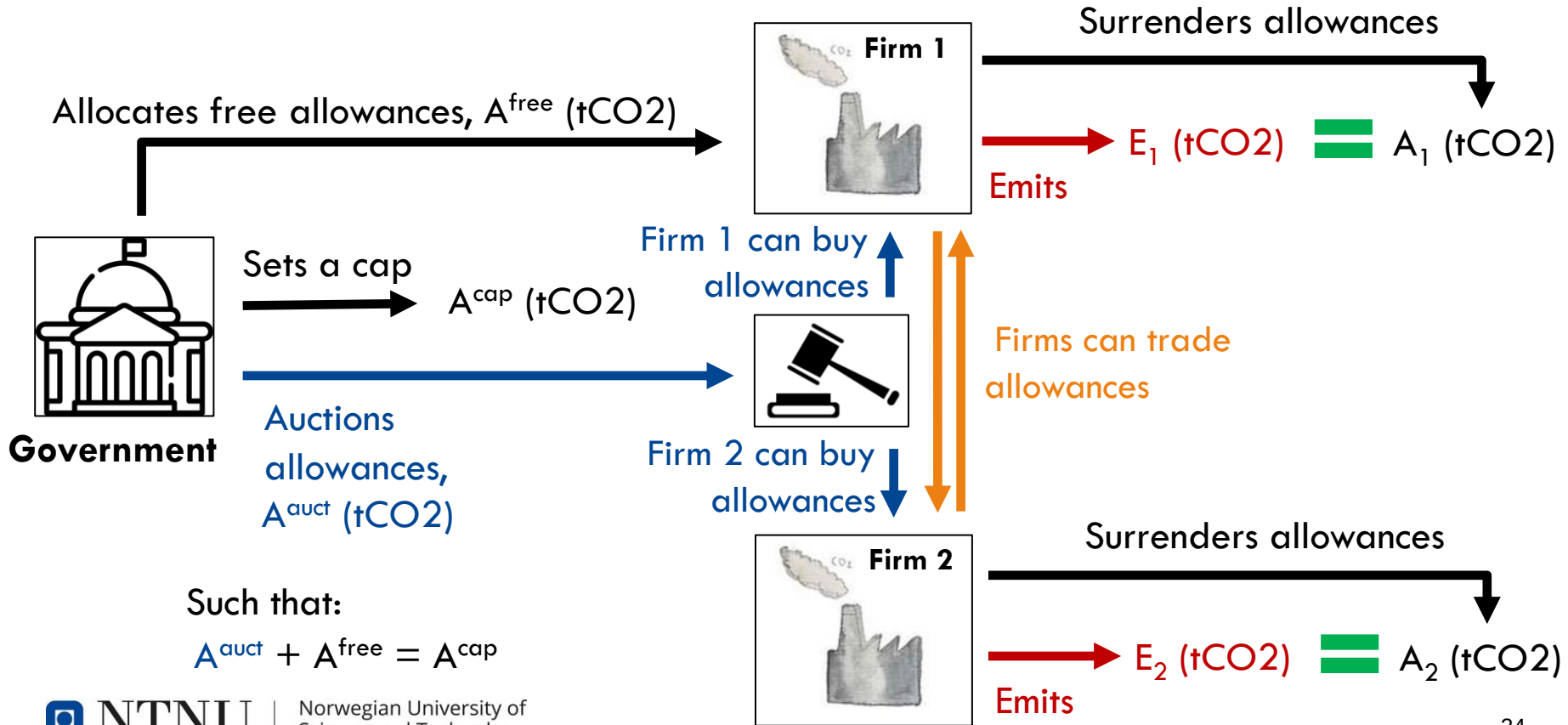
The fundamentals of how cap-and-trade works

1. Governments decide on the total quantity that can be emitted in a given year (the **cap**, in tons of CO₂)
2. **Allowances**¹ are made available² to companies.
 - Each allowance represents the right to emit 1 ton CO₂-eq
 - The total number of allowances equals the cap
3. Emitting companies are required to **surrender** one allowance¹ for every ton of CO₂-eq emitted at regular intervals of time (e.g., every year)

¹ These are called European Union Allowances (EUAs) in the EU

² For free and/or auctioned off

The fundamentals of how cap-and-trade works



Modeling cap-and-trade

The commonly used way to model cap-and-trade (emissions trading) in optimization:

$$\begin{aligned} \max_{p_g^G, p_d^D} \quad & \sum_d^{N_D} U_d p_d^D - \sum_g^{N_G} C_g p_g^G \\ \text{s.t.} \quad & \sum_d^{N_D} p_d^D - \sum_g^{N_G} p_g^G = 0 \quad (\lambda) \quad \leftarrow \text{Power balance} \\ & p_g^G \leq \bar{P}_g^G \quad (\mu_g) \quad \forall \quad g \quad \leftarrow \text{Capacity limit} \\ & p_d^D \leq \bar{P}_d^D \quad (\gamma_d) \quad \forall \quad d \end{aligned}$$

Variables:

p_g^G : generation of technology g (MWh)

p_d^D : demand of bid g (MWh)

Parameters:

C_g^G : Variable cost of technology g (\$/MWh)

U_d^D : Utility (i.e. maximum willingness to pay)
for bid d (\$/MWh)

$\bar{P}_g^G$: Capacity of technology g (MW)

Modeling cap-and-trade

The commonly used way to model cap-and-trade (emissions trading) in optimization:

$$\begin{aligned} \max_{p_g^G, p_d^D} \quad & \sum_d^{N_D} U_d p_d^D - \sum_g^{N_G} C_g p_g^G \\ \text{s.t.} \quad & \sum_d^{N_D} p_d^D - \sum_g^{N_G} p_g^G = 0 \quad (\lambda) \end{aligned}$$

← Power balance

$$p_g^G \leq \bar{P}_g^G \quad (\mu_g) \quad \forall \quad g$$

← Capacity limit

$$p_d^D \leq \bar{P}_d^D \quad (\gamma_d) \quad \forall \quad d$$

Cap on CO₂ →

$$\sum_g E_g^{CO_2} p_g^G \leq \bar{E}^{CO_2} \quad (\kappa)$$

↑ CO₂ price

Variables:

p_g^G : generation of technology g (MWh)

p_d^D : demand of bid g (MWh)

Parameters:

C_g^G : Variable cost of technology g (\$/MWh)

U_d^D : Utility (i.e. maximum willingness to pay) for bid d (\$/MWh)

$\bar{P}_g^G$: Capacity of technology g (MW)

$E_g^{CO_2}$: Emissions intensity of technology g (tCO₂/MWh)

$\bar{E}^{CO_2}$: Cap on emissions (tCO₂)



Time for an exercise

How does the cap on CO₂ impact the electricity price?

How to find out: derive the stationarity KKT condition with respect to generation

Steps

1. Derive the Lagrangian of the problem
2. Differentiate the Lagrangian with respect to generation, p_g^G

The standard form of the model:

$$\text{Min}_{p_g^G \geq 0, p_d^D \geq 0} SW = \sum_{g=1}^{N_G} C_g p_g^G - \sum_{d=1}^{N_D} U_d p_d^D$$

$$\text{s.t.} \quad \sum_d p_d^D - \sum_g p_g^G = 0 \quad (\lambda)$$

$$p_g^G \leq \bar{P}_g^G \quad (\mu_g) \quad \forall \quad g$$

$$p_d^D \leq \bar{P}_d^D \quad (\gamma_d) \quad \forall \quad d$$

$$\sum_g E_g^{CO_2} p_g^G \leq \bar{E}^{CO_2} \quad (\kappa)$$

Solution

The stationarity condition with respect to generation:

$$C_g - \lambda + \mu_g + \kappa E_g^{CO_2} = 0 \quad \forall g$$

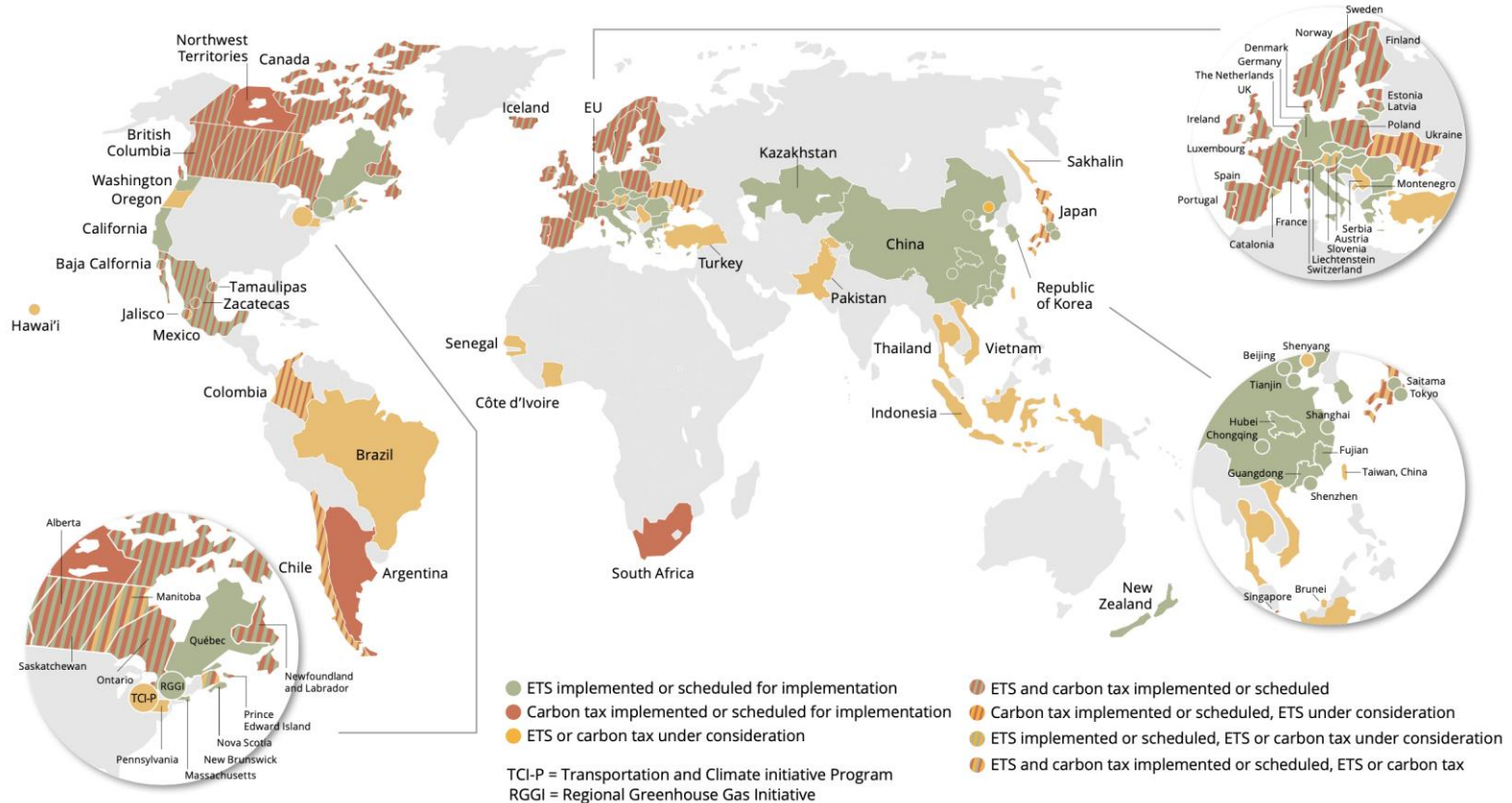
Now, let's rearrange:

$$\lambda = C_g + \mu_g + \kappa E_g^{CO_2} \quad \forall g$$

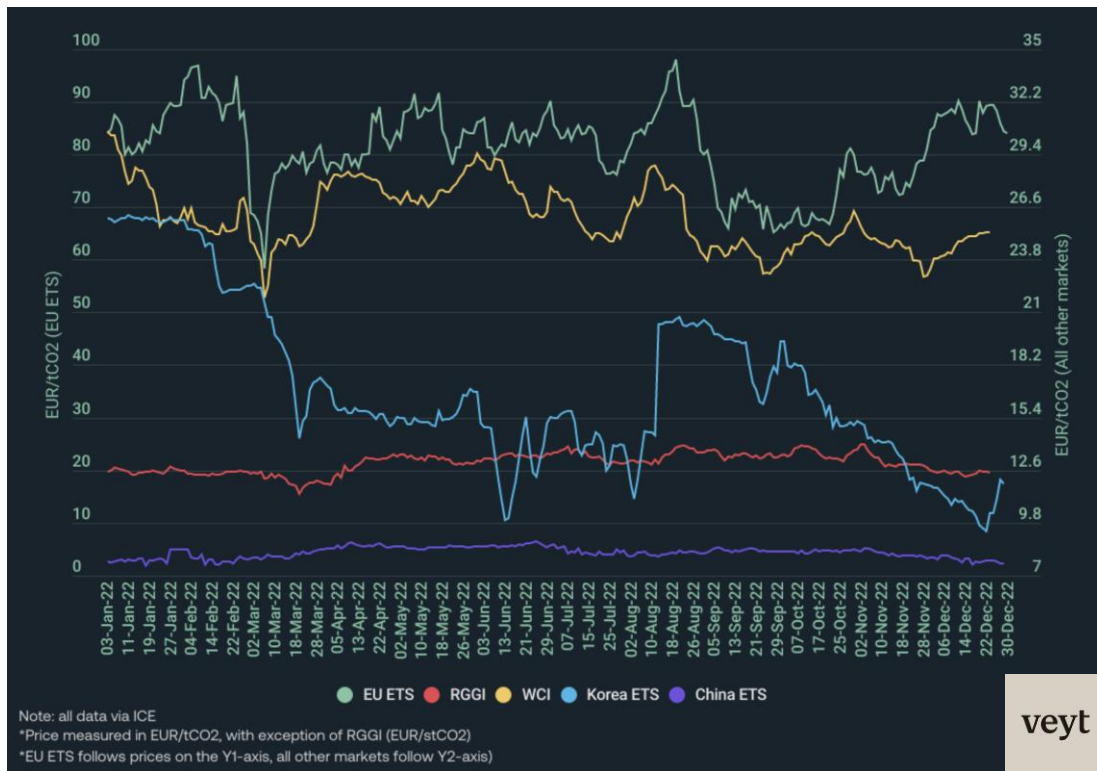
Observe that for the marginal g , $\mu_{g=m}$ is zero, so:

$$\lambda = C_m + \kappa E_m^{CO_2} \quad \dots \text{which is how we expect CO}_2 \text{ pricing to impact the power price (slide 24)}$$

Carbon pricing around the world



Carbon prices in different carbon markets



Source: Veyt. 2022. Oslo, Norway



The European Union Emissions Trading System

The European Emission Trading Scheme (EU ETS)^{1,2}

- EU Directive ("2003/87/EC")
 - First period: 2005 – 2007
 - Second period 2008 – 2012
 - Third period 2012 – 2020
 - Forth period 2021-2030 (annual reductions in allowances to 2.2% as of 2021)
- The world's largest "cap-and-trade" market
- Covers all EU countries plus Iceland, Liechtenstein and Norway
- Volume
 - 1.5 Gt CO₂ (40% emissions)
 - 4000 companies
 - 12 000 plants



¹ https://climate.ec.europa.eu/eu-action/eu-emissions-trading-system-eu-ets/about-eu-ets_en

² https://ec.europa.eu/clima/policies/ets_en

Carbon pricing in the context of EU climate policy

Current EU emissions ~ 4 Gt CO₂

EU Climate Law

Net zero GHG emissions by 2050 (legally binding target)

Energy sectors:

Electricity



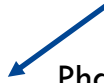
Industry



Aviation



Shipping



Ground transport



Buildings



EU ETS (now covers ~1.5 Gt)

Phased in
from 2024

ETS 2 set to launch in 2027
CO₂ price cap of €45/t

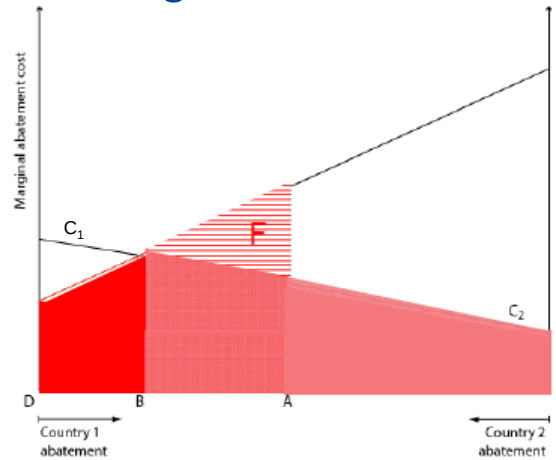
CO₂ pricing only one instrument of many:

Renewable Energy Directive

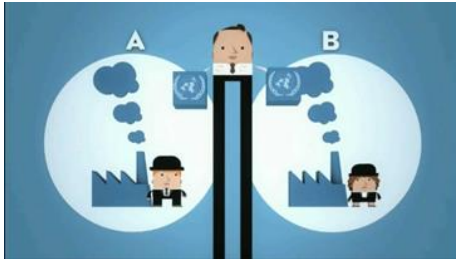
Energy Efficiency Directive

Nation-specific laws

Reducing Abatement Costs Through Emissions Trading



DE - Total abatement
A - Equal abatement
B - Equal marginal cost = cost minimization
F - Cost saving potential



Trade reduces abatement cost

- C_1 and C_2 marginal abatement costs for Country 1 and 2, respectively
- A: required abatement
- B: Optimal abatement
- Total cost without trade equals the sum of shade and the hatched area
- If Country 1 buys abatement in Country 2, total costs are reduced with size of area **F**

ICAP Allowance Price Explorer



EU ETS: key design features

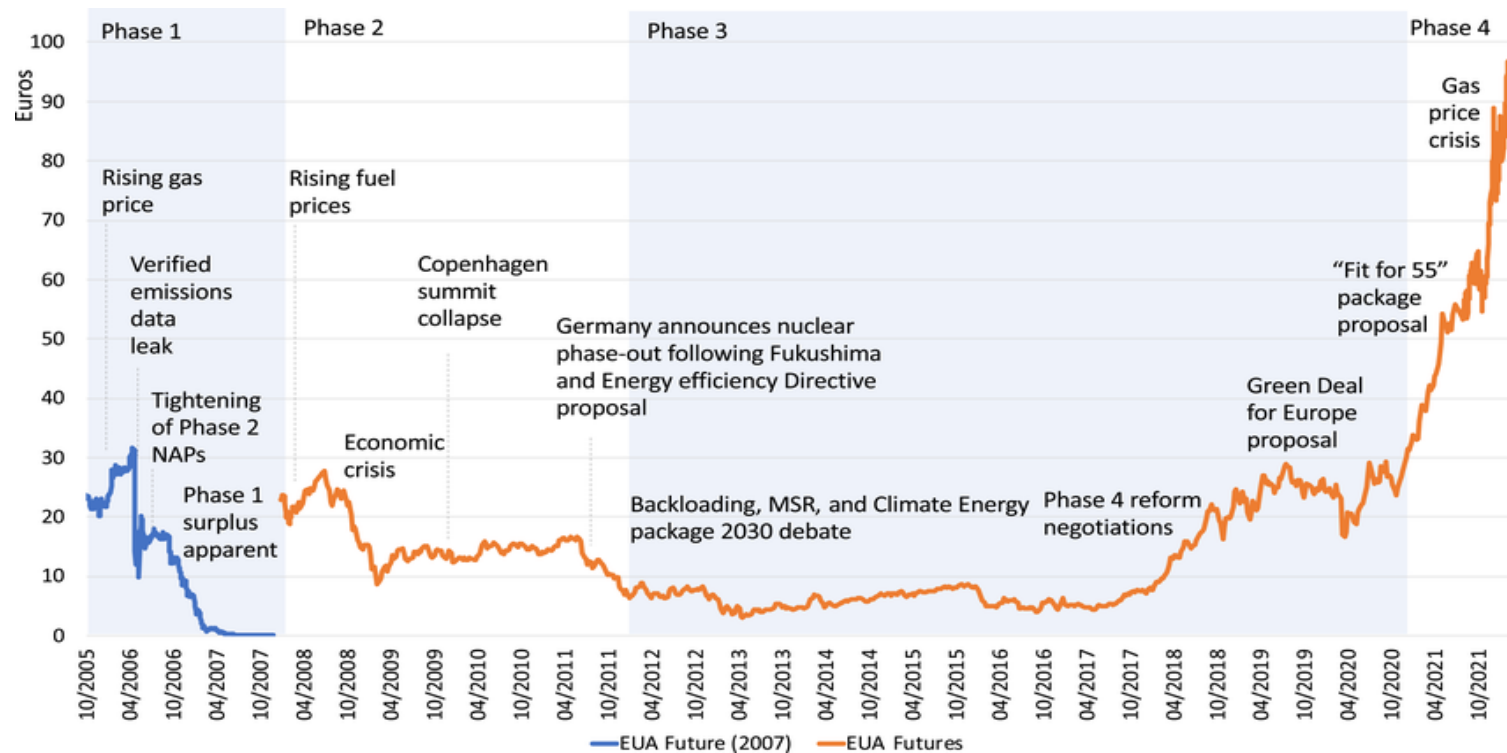
- Scope
 - Sectors: power, industry, aviation, ~ 1.5 Gt CO₂
 - GHGs: CO₂, N₂O, PFCs
- Declining cap: -62%¹ by 2030 (reaches 100% in 2039²)
- Allowances are auctioned and freely allocated
- Revenues used for, e.g., clean energy innovation



Key issues

- How ambitious should the cap be?
- Economic impacts, especially industrial competitiveness and **carbon leakage**
- **Carbon leakage**: When emission *reductions in a regulated country* result in emission *increases in an unregulated trade partner country*
- How to avoid an oversupply of allowances: **Market Stability Reserve (MSR)** adopted in 2015

Prices and Major Events



Technology policies for clean energy sources

The purpose of clean energy policy

- Economics 101 says climate change is an externality, and a sufficient climate policy would be to “internalize” the externality via a CO₂ price.

However, there are reasons to support clean energy:

- **Political constraints** prevent implementation of optimal CO₂ prices¹
- **Market failures** *other than* the CO₂ externality make the economy sub-optimal. Clean energy policy can increase welfare.²
 - Innovation related **knowledge spill-overs** are an important market failure that prevents optimal investment in new technology (**R&D** and **learning-by-doing** in the renewables industry)
 - **Energy security** effects (countries often seek to decrease reliance on imported fuels)
 - Cost of new technology will probably decrease too slowly

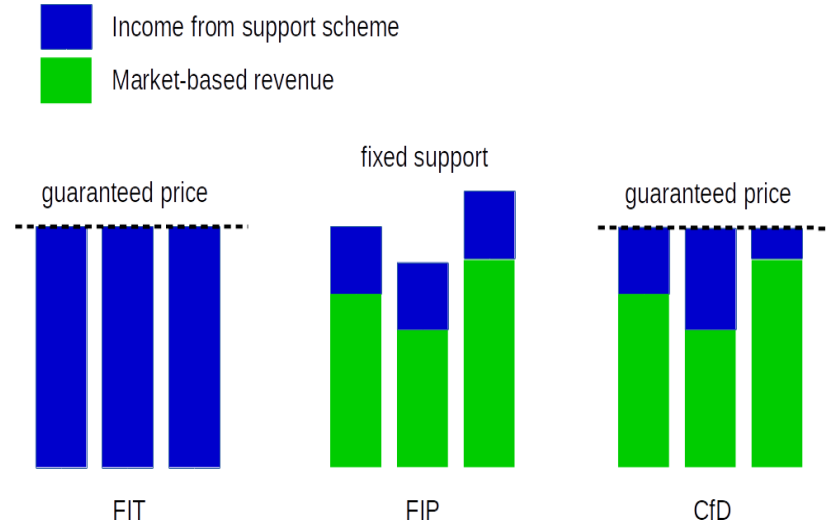
¹ relevant theoretical foundations: [Lipsey and Lancaster \(1956\)](#)

² e.g., see Lehman and Gawel 2013. “Why should support schemes for renewable electricity complement the EU emissions trading scheme?”. *Energy Policy* 52.

Price-based renewable policy

The 3 most common support schemes are:

- ☐ *feed-in-tariff* (FIT)
- ☐ *fixed feed-in premium* (FIP)
- ☐ *contract for difference* (CfD, or sliding premium)



These may have an impact on participant revenues, offering strategies and market outcomes

Price-based renewable policy, continued

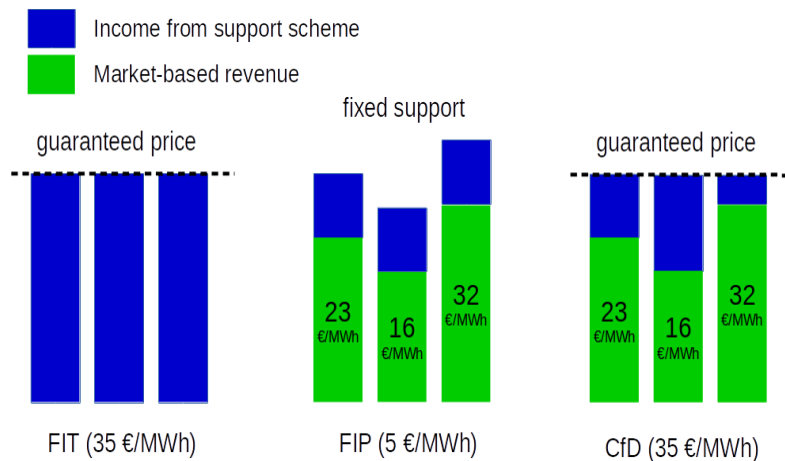
- Let us consider the case of a renewable energy producer (wind or solar) participating in the day-ahead electricity market

- 3 cases:

- FIT at 35 €/MWh
- FIP of 5 €/MWh
- CfD to guarantee 35 €/MWh

- Revenues are:

- FIT case: $35 \times 3 = 105 \text{ €}$
- FIP case: $(23+16+32) + 5 \times 3 = 96 \text{ €}$
- CfD case: $(23+16+32) + (12+19+3) = 105 \text{ €}$

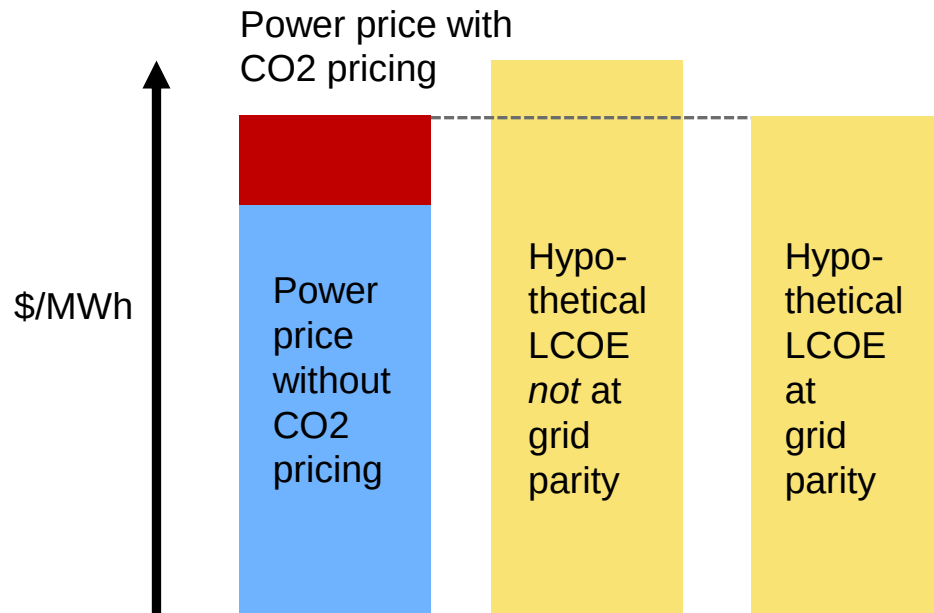


Key concepts for understanding the economics of renewables

Levelized Cost of Energy (LCOE): the total cost of generating electricity¹; Indicates the market price over the technology lifetime that would be necessary for the technology to break even.

- Meant for project finance purposes
- Can also be used for studying the evolution of a technology's costs over time
- Not meant for comparing different technologies

Grid parity: when the LCOE equals the electricity market price



¹ includes capital costs, O&M, taxes, performance characteristics, and variable costs.

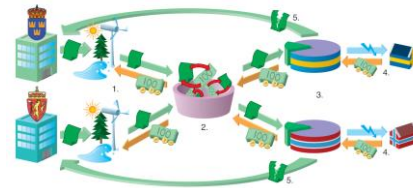
Fundamentals of Tradable performance standards in electricity

- Purpose is to increase renewable/clean generation
- Known as **Green certificate schemes** in Europe,
- Called **Renewable portfolio standard** (RPS) or **Clean Energy Standards** (CES) in the U.S.
- These policies revolve around the central new concept of the “**green certificate**” (a.k.a., “clean energy **credit**”).
- The **certificate** grants the owner the property right to the environmental attributes of the (renewable/clean) electricity being generated
 - Why do we need them? Since electrons on the grid are indistinguishable, we need a separate way of keeping track of who is consuming clean electricity and how much

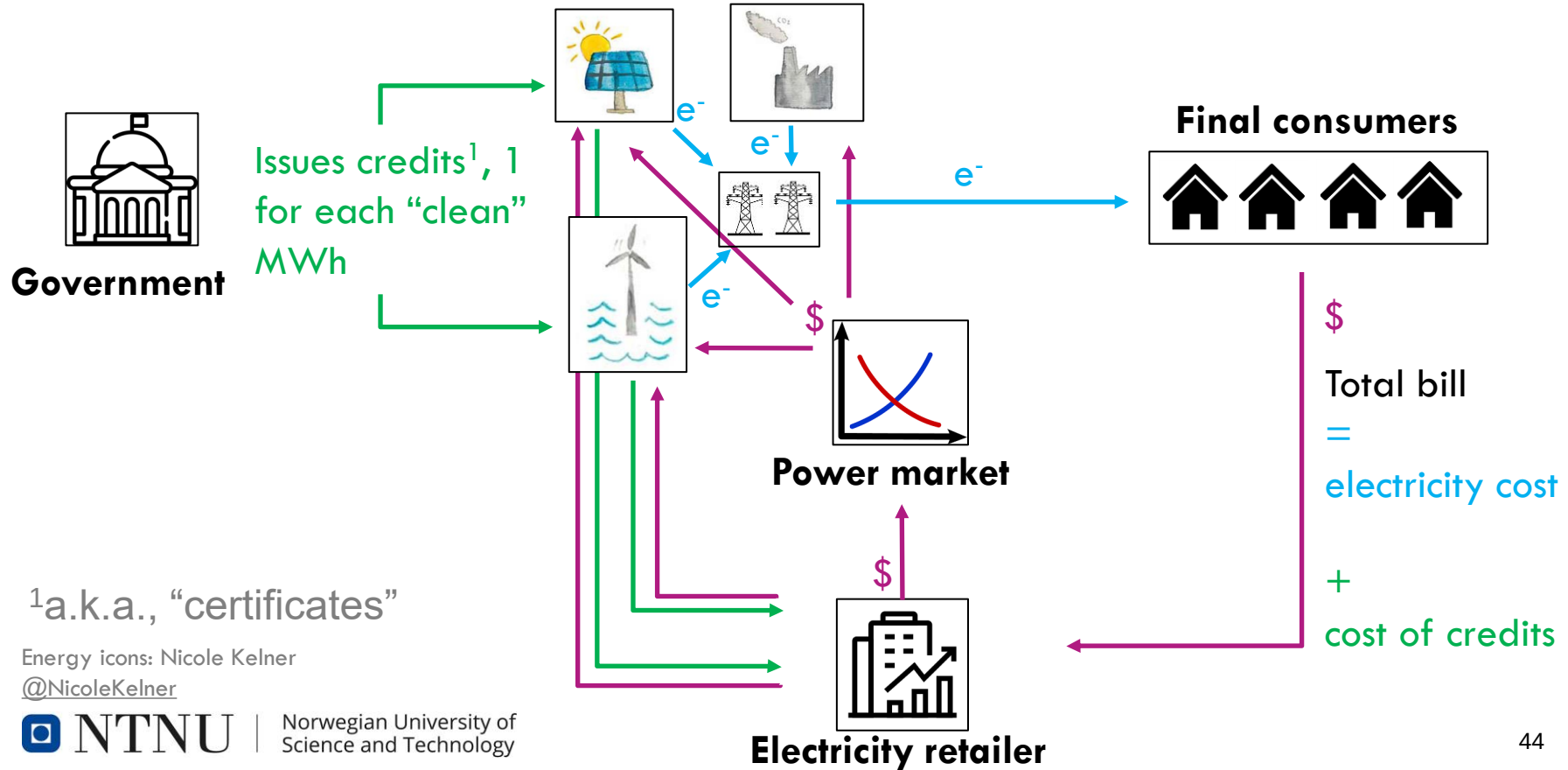
Tradable Green Certificates (TGC)

- Electricity certificates are a technology-neutral support scheme that makes it more profitable to invest in power production based on renewable energy sources (water, wind, solar and bioenergy).
- From 1 January 2012, Norway became part of a common electricity certificate market with Sweden.
- The countries had a common goal of 28.4 TWh of new renewable power production by the end of 2020.
- Power plants that are approved in the scheme were awarded electricity certificates for up to 15 years.
- Norwegian power plants approved in the scheme had to put into operation by the end of 2021.
- In Sweden, it has been decided to continue the scheme with a new national target between 2020 and 2030.

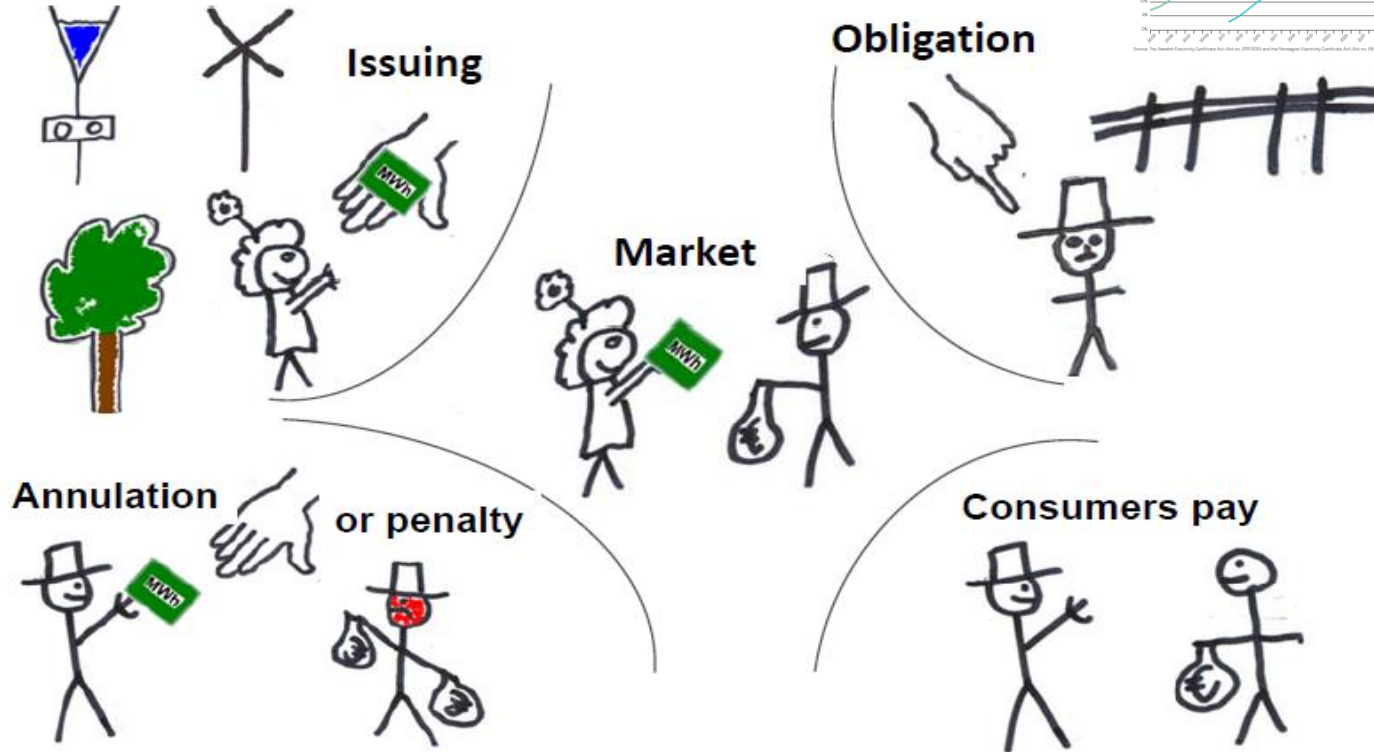
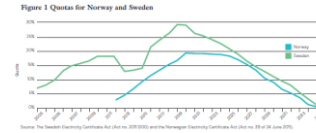
Source: <https://www.nve.no/energiforsyning/elsertifikater/>
https://www.nve.no/Media/4750/elcertifikat-2015-en_web.pdf



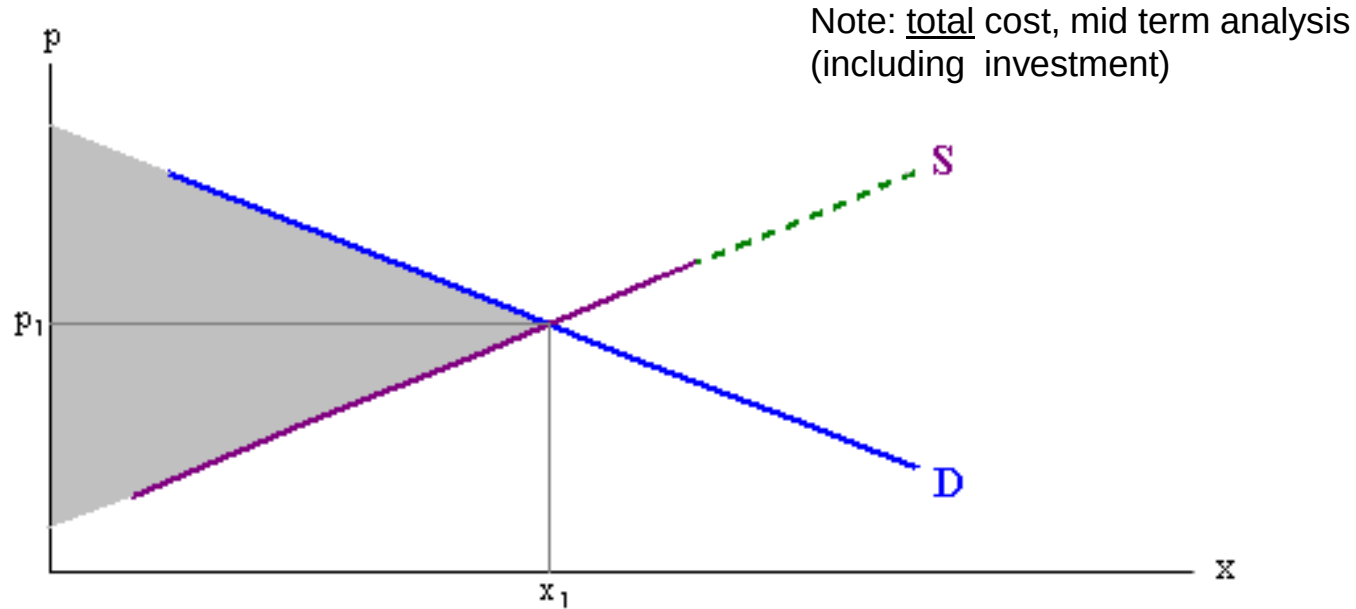
How Green Certificate schemes (a.k.a., Clean energy standards) work



Green certificate system

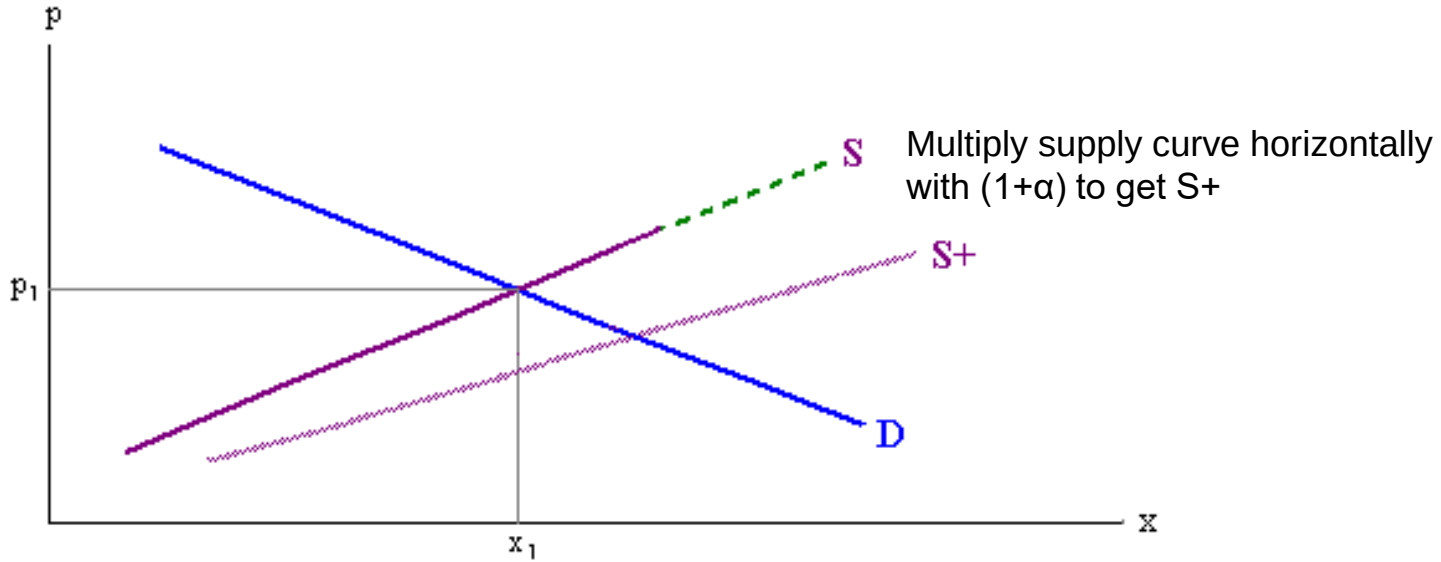


Economic consequences of TGC (1)



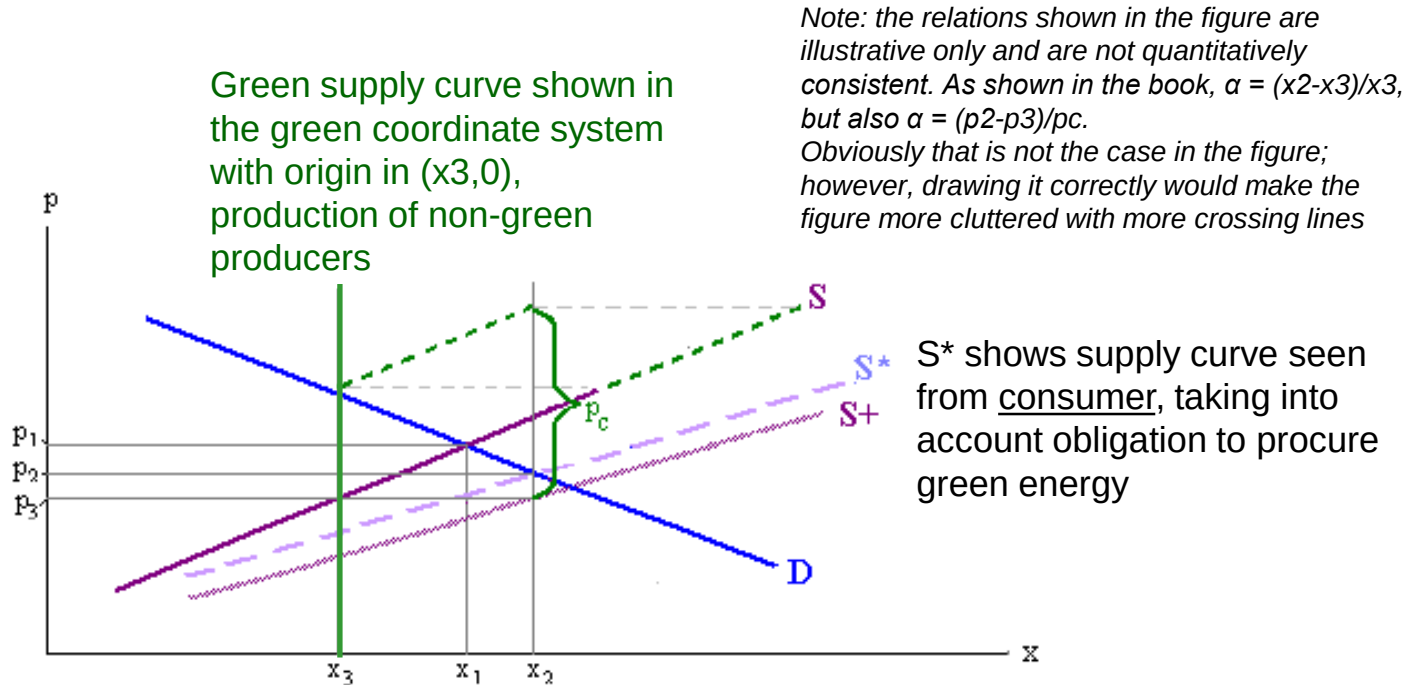
The Equilibrium point (x_1, p_1) between the supply (S) and demand (D) curve. The upper dotted part of the supply curve stands for the renewable electricity supply. P_1 is the market price and x_1 is the amount of electricity that can be sold at the market price or less. The socio-economic surplus is indicated as the grey shaded area.

Economic consequences of TGC (2)



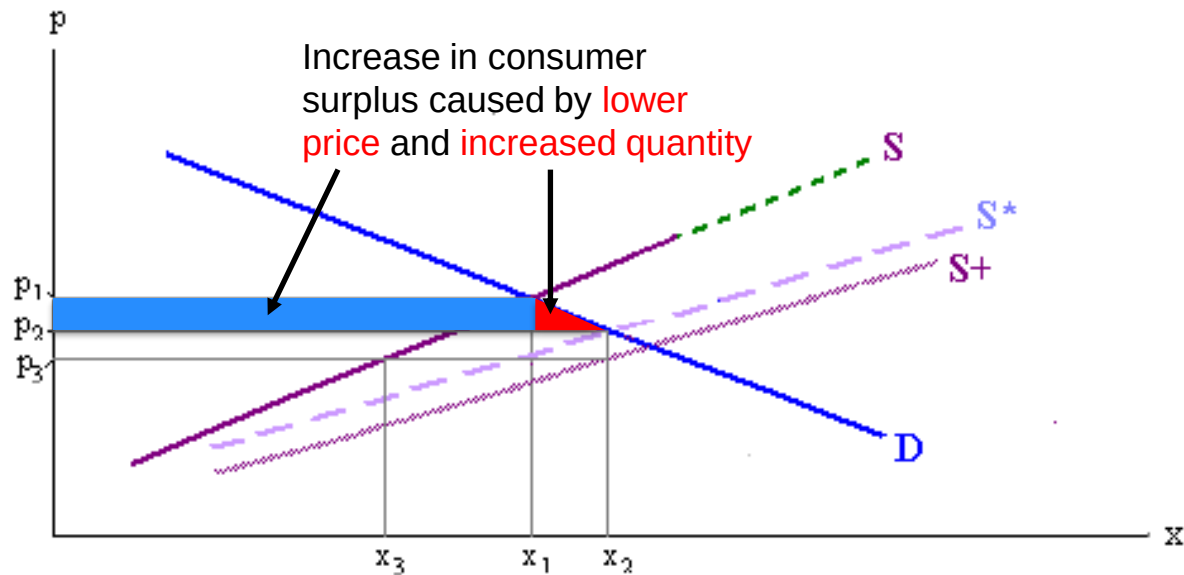
Integrating renewable generation into the supply. $S+$ comprises the new supply curve, but does not include the additional cost that the renewable share will inflict.

Economic consequences of TGC (3)



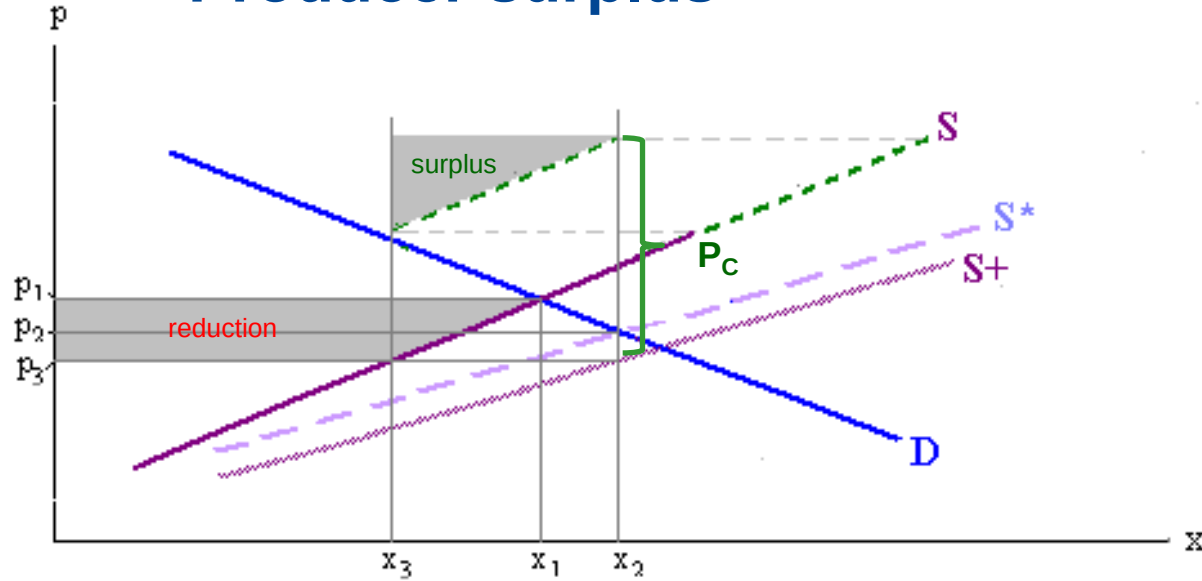
When including the additional consumer's share of green electricity, α , an additional cost from the purchase of certificates is added. This supply curve to shift upwards, corresponding to α^* represented by S^* . p_c equals the TGC price.

Consumer surplus



The increase in consumer's surplus, indicated by the grey shaded area

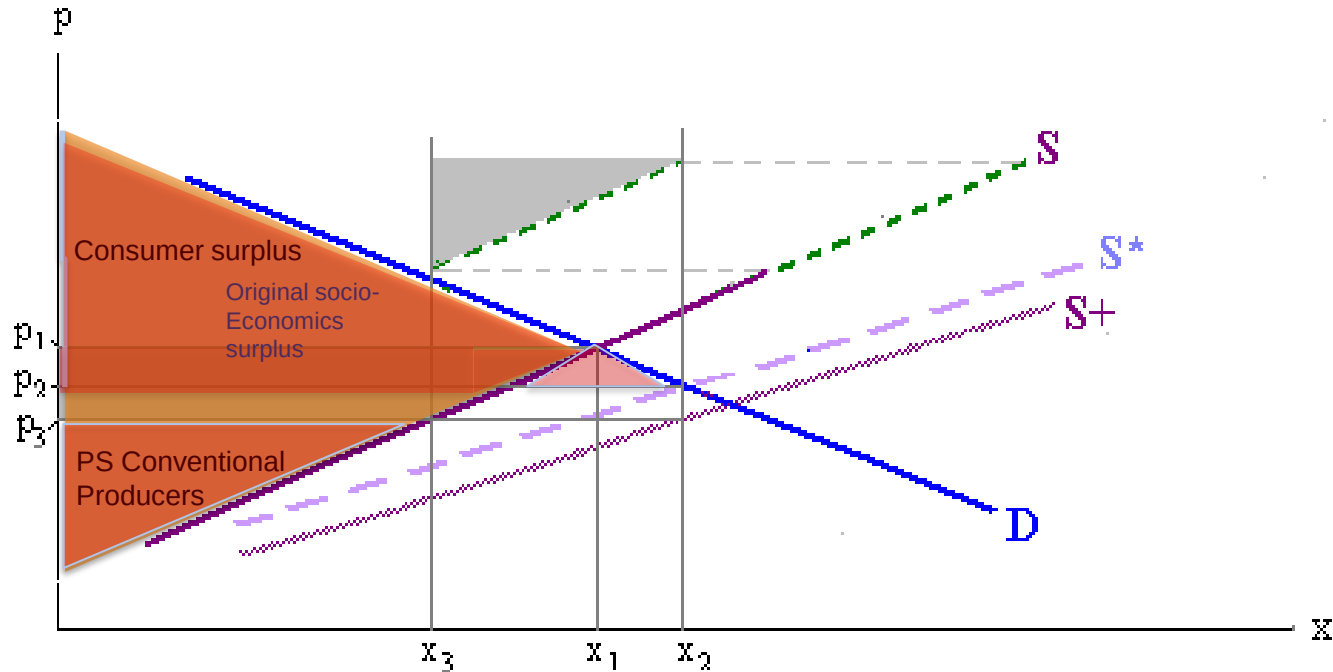
Producer surplus



Producers of **conventional** electricity experience a **reduction** in surplus represented by the lower shaded grey area to the left of the supply curve (S), between p_1 and p_3 (lower price, reduced volume)

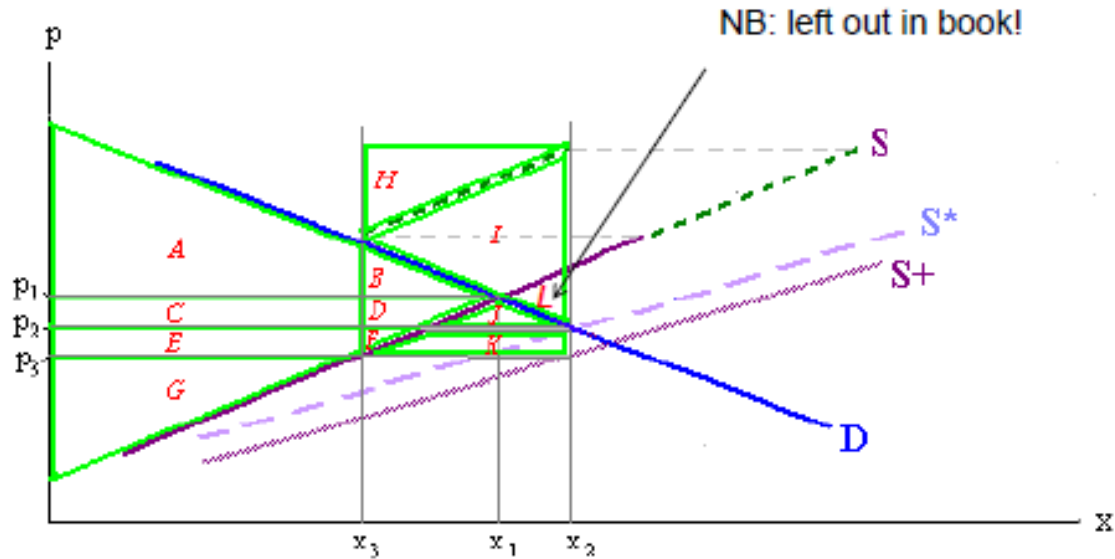
Renewable electricity producers receive a **surplus** corresponding to the upper shaded grey triangular area.

Economic consequences of TGC (4)



Total socio-economic surplus after including a green electricity share on the electricity supply

Socio-economic surplus with TGC market



Prior to the introduction of the TGC:

Consumer's surplus = A + B

Producer's surplus = C + D + E + F + G

Socio-economic surplus = A + B + C + D + E + F + G

After the introduction of the TGC:

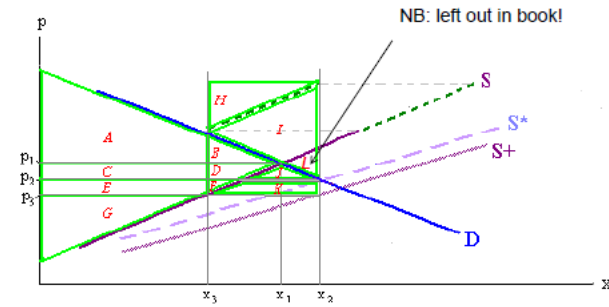
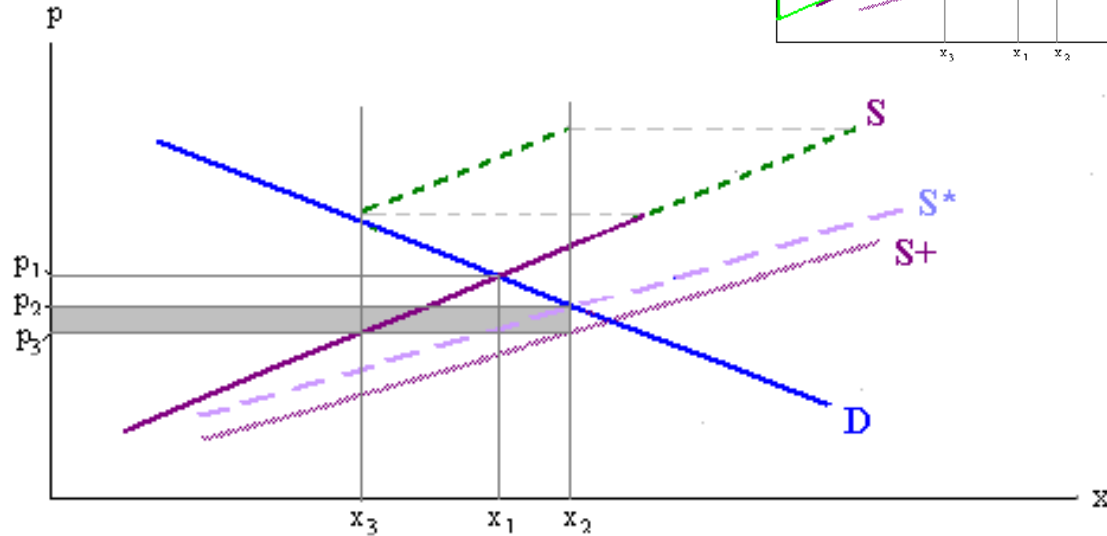
Consumer's surplus = A + B + C + D + J

Conventional Producer's surplus = G

Green Producer's surplus = H

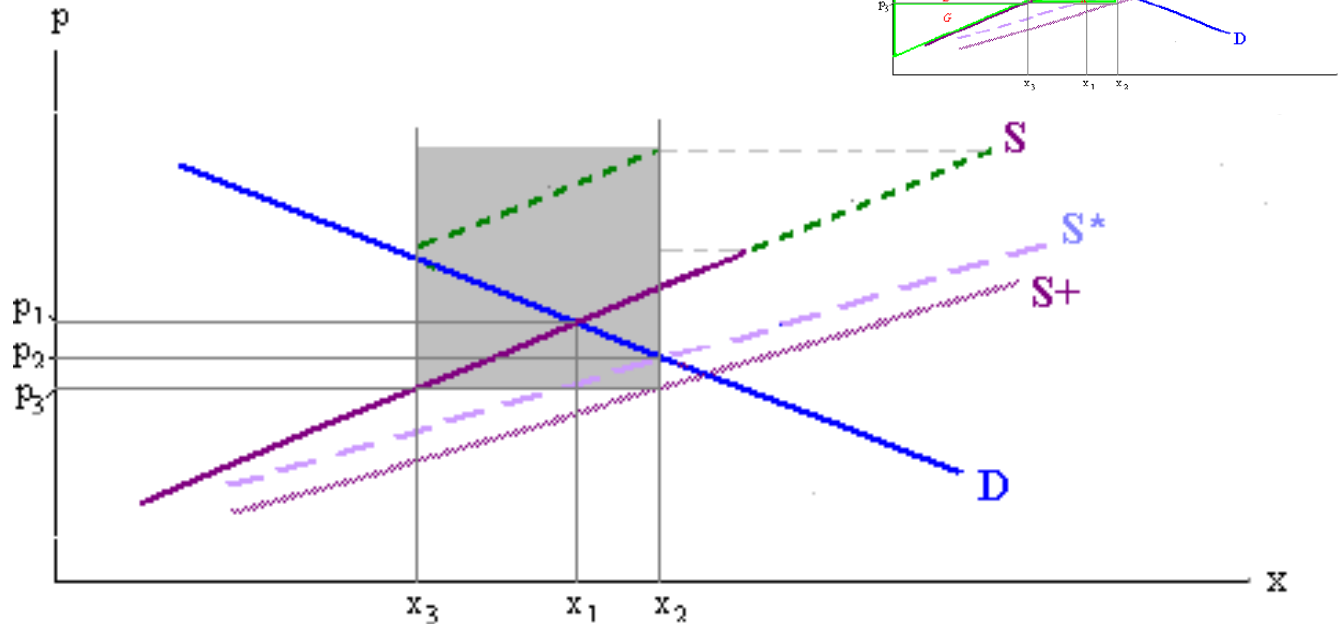
Socio-economic surplus = A + B + C + D + J + G + H

The consumer's cost



The consumer's cost for purchasing renewable certificates, shown as the grey shaded area
 $= E + F + K$

The renewable electricity producers' income



The renewable electricity producers' income from selling green certificates, shown as the grey shaded area.

$$= H + I + B + D + J + F + K + L$$

TGC: Loss in economic surplus

- Cost of purchasing renewable certificates = Income from selling renewable certificates

$$\begin{aligned} E + F + K &= H + I + B + D + J + F + K + L \\ \Rightarrow (J + H) &= E - I - B - D - L \end{aligned} \quad (1)$$

- Change in socio-economic surplus (ES)**

$\Delta ES = \text{ES with TGC market} - \text{ES without TGC market}$

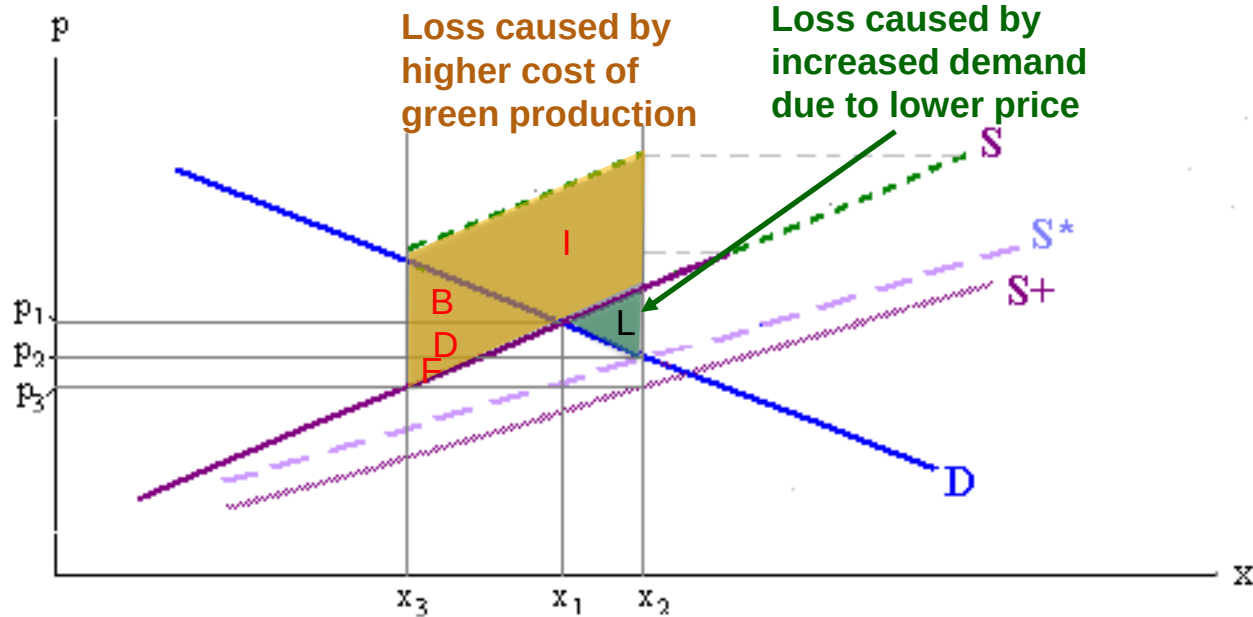
$$= (A + B + C + D + J + G + H) - (A + B + C + D + E + F + G)$$

$$= J + H - E - F \stackrel{(1)}{\Rightarrow} (E - I - B - D - L) - E - F$$

$$= -I - B - D - F - L$$

- Since all of the components are **negative**, the change in socio-economic surplus is bound to be negative

TGC: Loss in economic surplus



- The loss economic surplus, shown as the shaded area, due to the implementation of the TGC market.
- Note: we did not take into account the **externalities**